Week 11

Real-World Multi-Omics Case Studies

From Research Papers to Biological Discovery

Interpreting genomics, transcriptomics, microbiome, metabolomics, proteomics and single-cell data through original published figures and cross-omics logic.



Paper selection
question • design • cohort

→

Original figures
published evidence

→

Data interpretation
statistics • biology • caveats

→

Cross-omics integration
linking mechanisms

→

Scientific insight
from evidence to hypothesis

Two 40-Minute Sessions | Fall 2026
Bioinformatics: From Multi-Omics Data to Discovery

Dr. Liwei Xie | Shenzhen University of Advanced Technology

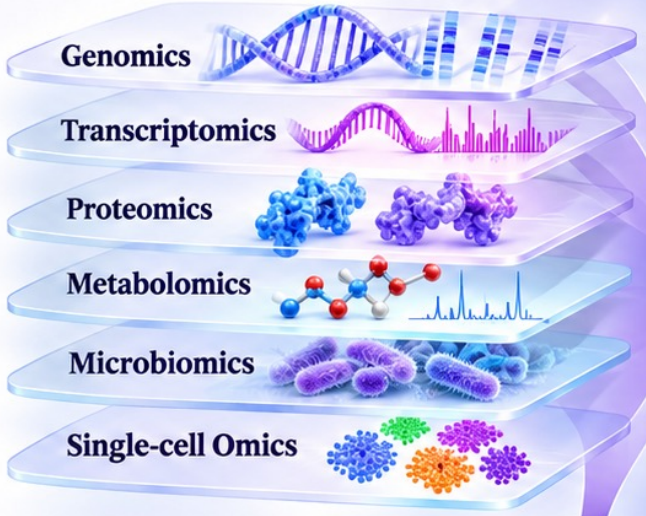
Week 11 | Bioinformatics: Multi-Omics Data Analysis

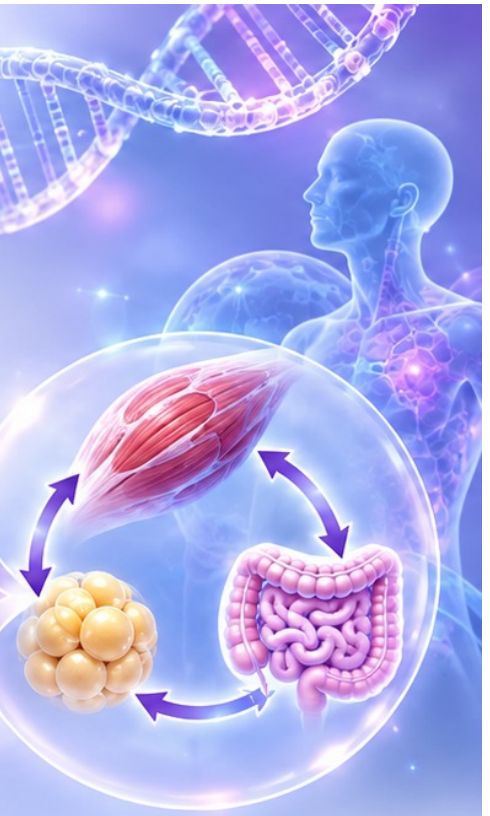
Multi-Omics Case Studies

From Experimental Design to Data Interpretation and Integration

Outline

1. Multi-Omics Study Workflow
2. Step 1: Study Design & Question
3. Step 2: Data Processing & QC
4. Step 3: Differential Analysis & Functional Annotation
5. Step 4: Cross-Omics Integration
6. Step 5: Biological Interpretation & Validation
7. Case Studies from Top Journals
8. Key Takeaways





Cross-Omics Integration

Multi-omics Integration → Key Signatures & Pathways → Mechanistic Insights → Intervention & Translation

Week 11 | Bioinformatics: Multi-Omics Data Analysis

Step 1. Paper Selection

How to choose a high-quality multi-omics study for in-depth analysis?

Genomics
Transcriptomics
Proteomics
Metabolomics
Microbiomics

Selection Criteria

- ☑ **Clear biological question and hypothesis**
- ☑ **Well-designed cohort or experimental model**
- ☑ **Multi-omics with proper sample matching**
- ☑ **Key results with interpretable figures and validations**
- ☑ **High-impact journals** (Nature Metabolism, Nature Microbiology, JCI, etc.)



Week 11 | Bioinformatics: Multi-Omics Data Analysis

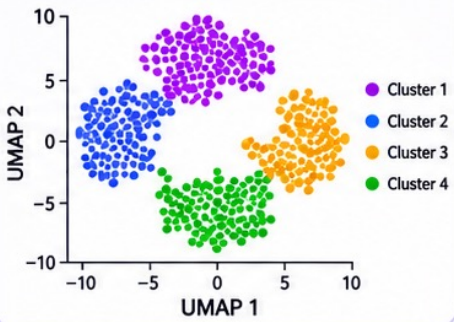
Step 2. Original Figures

Read the published figures: what do they show and how are they generated?

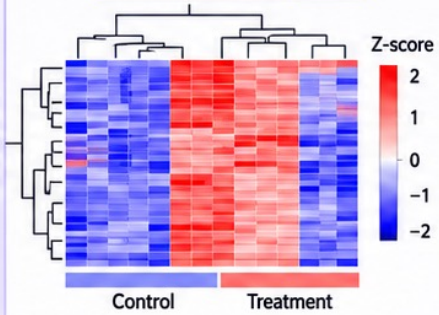
Key Questions

- What is the **main comparison** or **hypothesis**?
- What do the **axes, colors and symbols** represent?
- What **statistical test** and **significance** are used?
- What is the **biological message** of this figure?

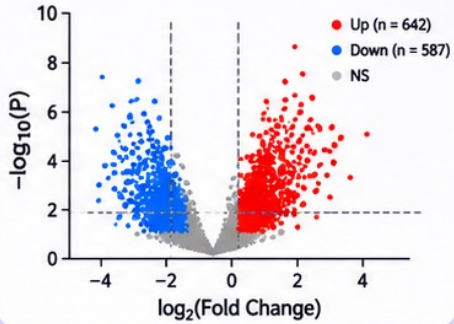
UMAP



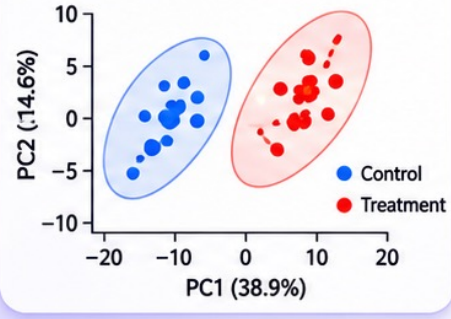
Heatmap



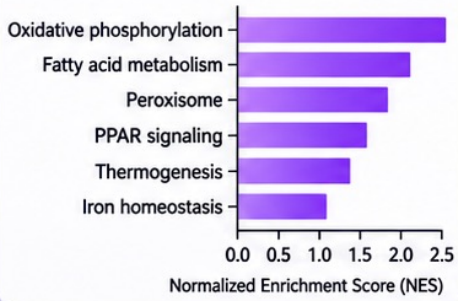
Volcano plot



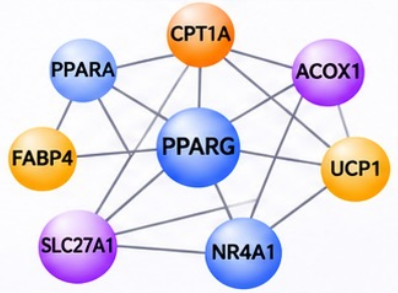
PCA



Pathway (GSEA)



Network



Week 11 | Bioinformatics: Multi-Omics Data Analysis

Step 3. Data Interpretation

From statistical results to biological meaning

Interpretation Framework

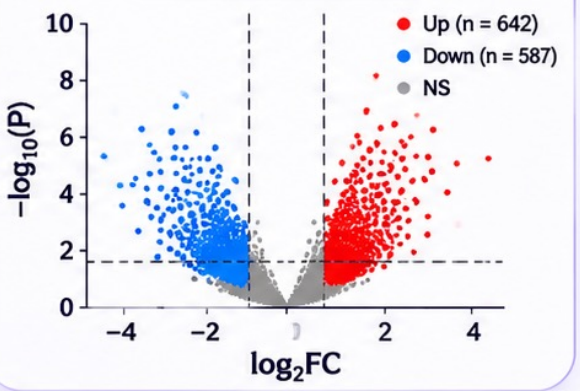
Data quality & reliability
(QC, normalisation, batch effect)

Key results
(DEG/DEP, abundance changes)

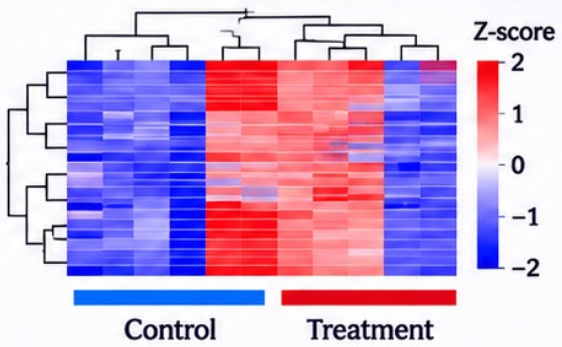
Functional annotation
(GO, KEGG, GSEA)

Biological inference
(what, how, and why?)

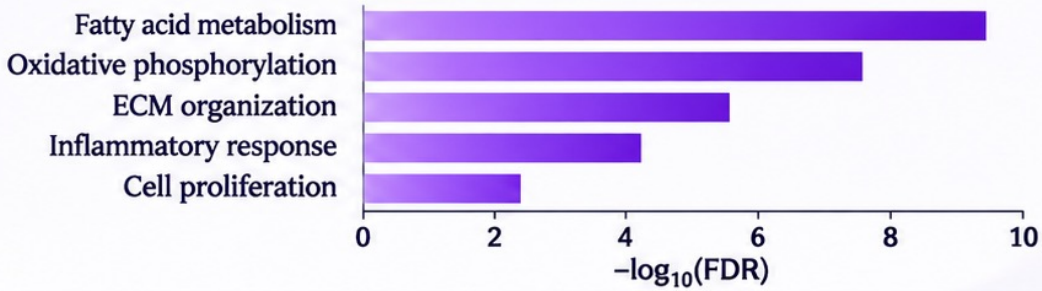
Volcano Plot



Heatmap



Pathway Enrichment (GSEA/ORA)



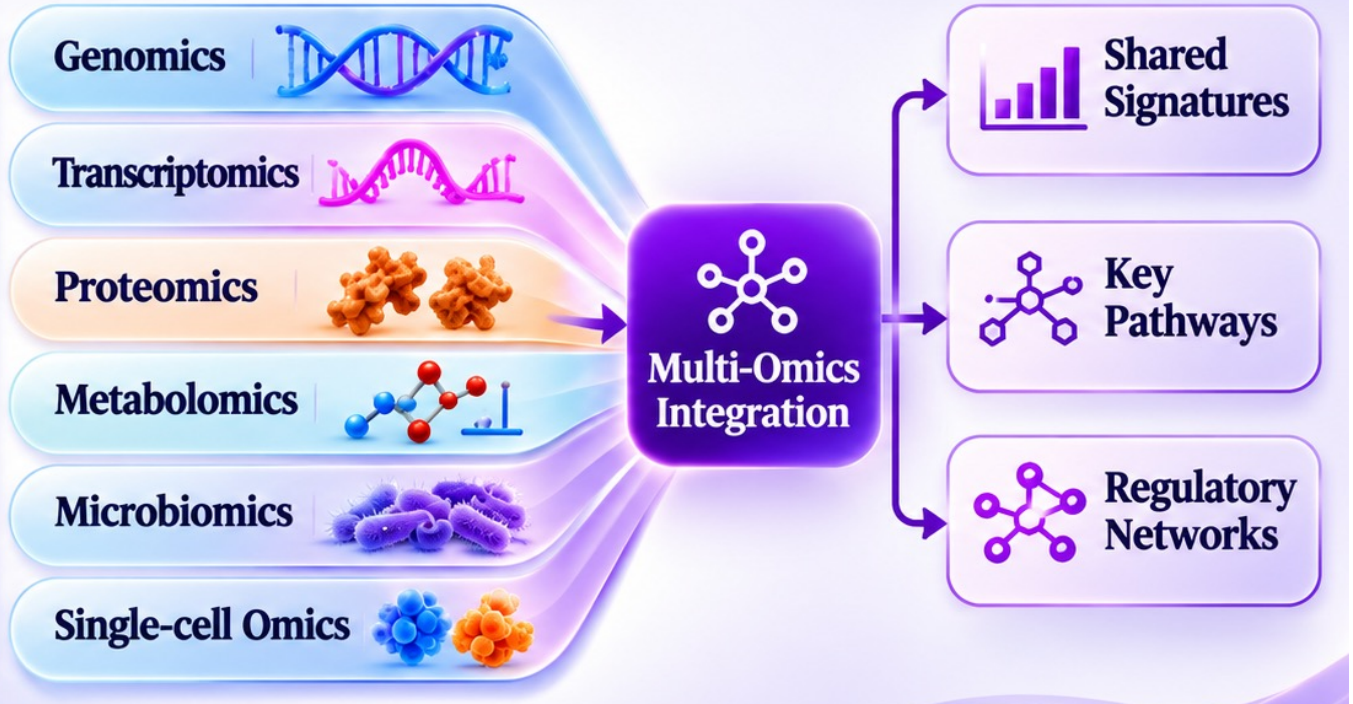
Week 11 | Bioinformatics: Multi-Omics Data Analysis

Step 4. Cross-Omics Integration

Integrate multi-omics data to identify key pathways and regulatory mechanisms

Integration Workflow

- **Data preparation** (matching, normalization)
- **Integration methods** (MOFA, DIABLO, sPLS, etc.)
- **Feature selection** (shared signatures)
- **Pathway & network analysis** (co-expression, regulation)



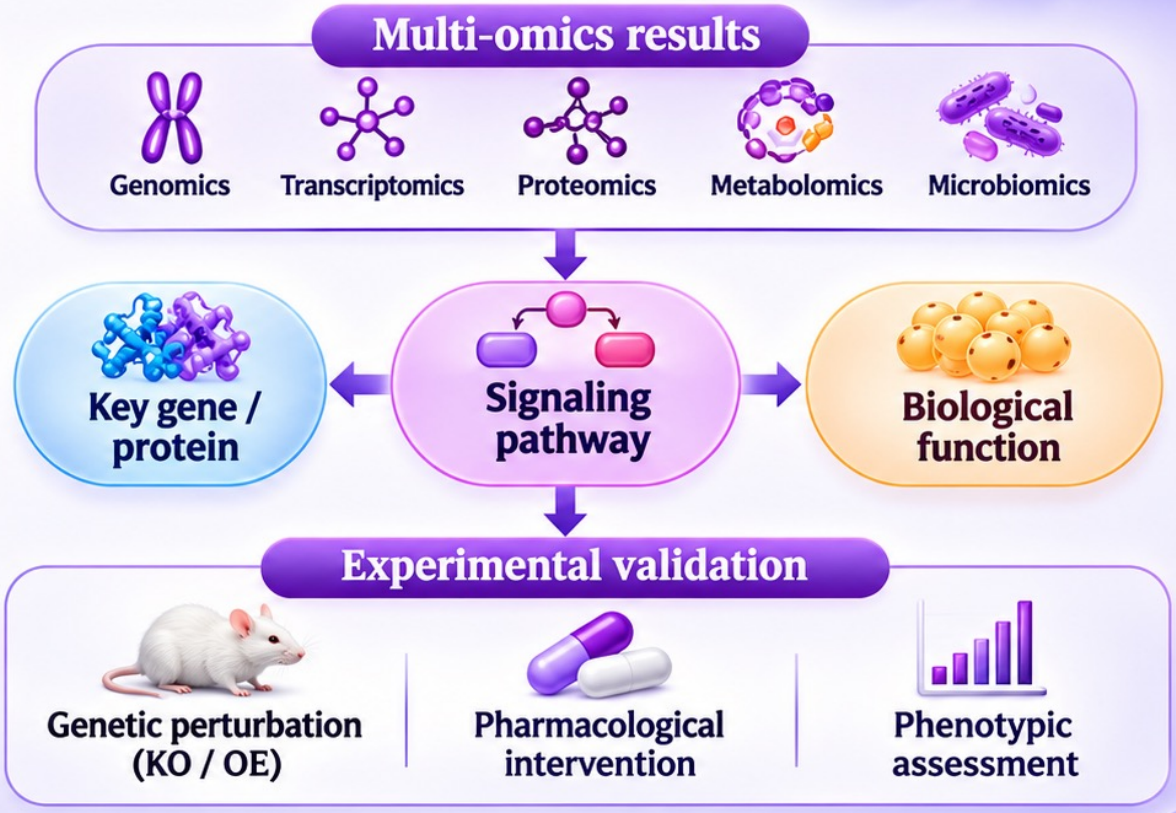
Week 11 | Bioinformatics: Multi-Omics Data Analysis

Step 5: Biological Interpretation & Validation

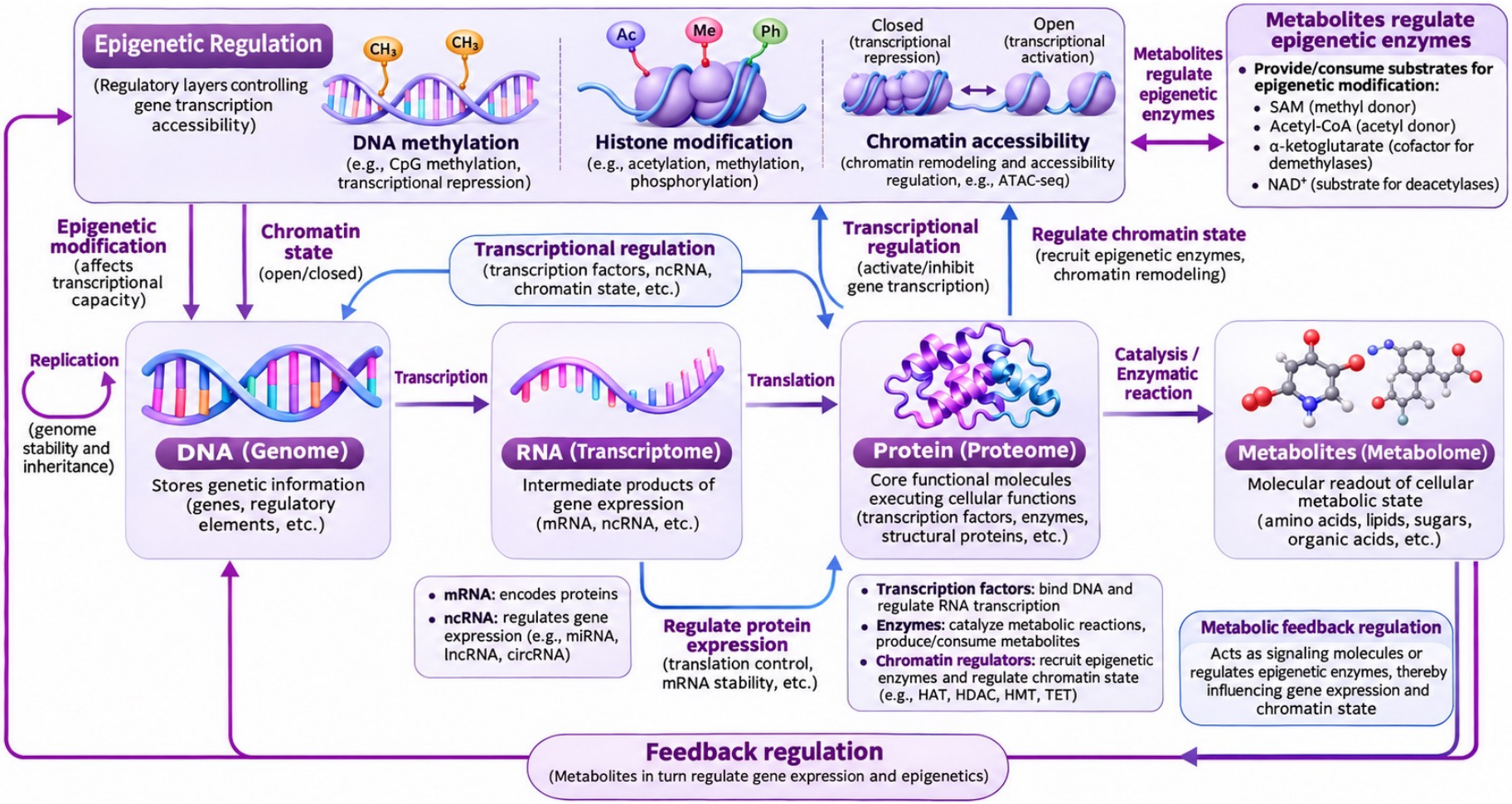
From data associations to mechanistic insights and experimental validation

Key Questions

- What is the **core biological finding?**
- Is it **correlation** or **causation?**
- Which **molecules** and **pathways** are key?
- How to **experimentally validate?**
- What are the **limitations** and **next steps?**



From Genome to Metabolome: An Integrated Multi-Omics Framework



Week 11 | Bioinformatics: Multi-Omics Data Analysis

Case Study 1: Proteomics

A Spatiotemporal Map of Human Adipogenesis

Klingelhuber et al. | *Nature Metabolism, 2024*



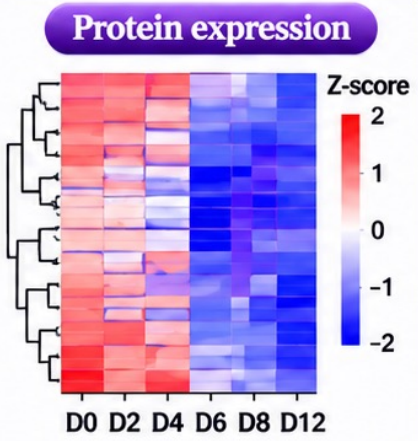
Study Overview

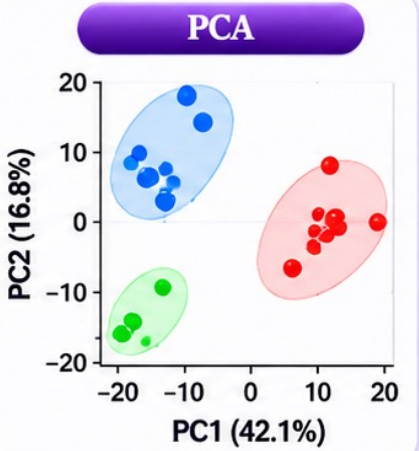
Question: How do protein networks change during human adipogenesis?

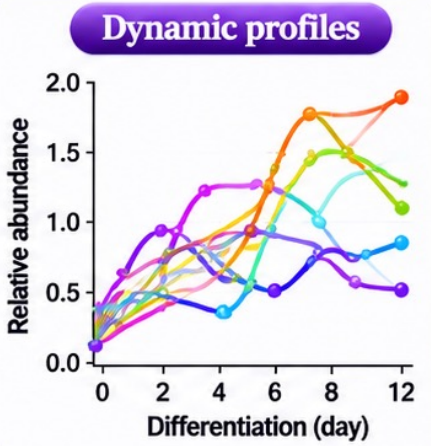
Approach: Time-resolved DIA proteomics

Key finding: Dynamic regulation of adipogenic and metabolic pathways







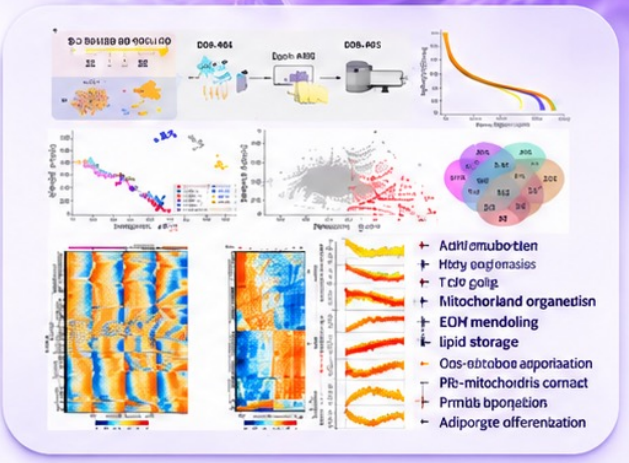


NATURE METABOLISM · 2024

Paper 1: Research Question

A spatiotemporal proteomic map of human adipogenesis





Klingelhuber et al. | 2024

ABSTRACT AT A GLANCE — paraphrased from the original



Protein maps capture temporal and organelle remodeling.



Metabolic enzymes change abundance and localization.



C19orf12 influences mitochondria-linked lipid storage.

Klingelhuber et al. | NATURE METABOLISM (2024) | https://doi.org/10.1038/s42255-024-01025-8

Abstract

nature metabolism

Article

https://doi.org/10.1038/s42255-024-01025-8

A spatiotemporal proteomic map of human adipogenesis

Received: 30 June 2023

Accepted: 7 March 2024

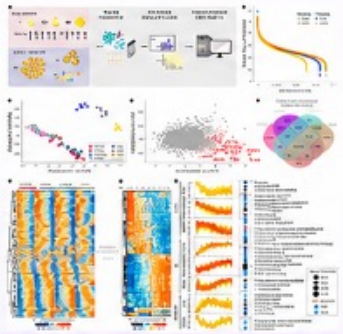
Published online: 2 April 2024

Check for updates

Felix Klingelhuber[1,2,11], Scott Frendo-Cumbo[3,11], Muhmmad Omar-Hmeadi[3], Lucas Massier[3], Pamela Kakimoto[1,2], Austin J. Taylor[1,2], Morgane Couchet[3], Sara Ribicic[1,2], Martin Wabitsch[4], Ana C. Messias[5,6], Arcangela Iuso[7,8], Timo D. Müller[1,2,9], Mikael Rydén[3,10], Niklas Mejhert[3] & Natalie Krahmer[1,2]

White adipocytes function as major energy reservoirs in humans by storing substantial amounts of triglycerides, and their dysfunction is associated with metabolic disorders; however, the mechanisms underlying cellular specialization during adipogenesis remain unknown. Here, we generate a spatiotemporal proteomic atlas of human adipogenesis, which elucidates cellular remodelling as well as the spatial reorganization of metabolic pathways to optimize cells for lipid accumulation and highlights the coordinated regulation of protein localization and abundance during adipocyte formation. We identify compartment-specific regulation of protein levels and localization changes of metabolic enzymes to reprogramme branched-chain amino acids and one-carbon metabolism to provide building blocks and reduction equivalents. Additionally, we identify C19orf12 as a differentiation-induced adipocyte lipid droplet protein that interacts with the translocase of the outer membrane complex of lipid droplet-associated mitochondria and regulates adipocyte lipid storage by determining the capacity of mitochondria to metabolize fatty acids. Overall, our study provides a comprehensive resource for understanding human adipogenesis and for future discoveries in the field.

Experimental Design



ORIGINAL PAPER FIGURE

SCIENTIFIC QUESTION

How do human adipocytes reorganize their proteins and organelles during differentiation?

Human models
4 adipocyte systems

Differentiate
Day 0 to 14

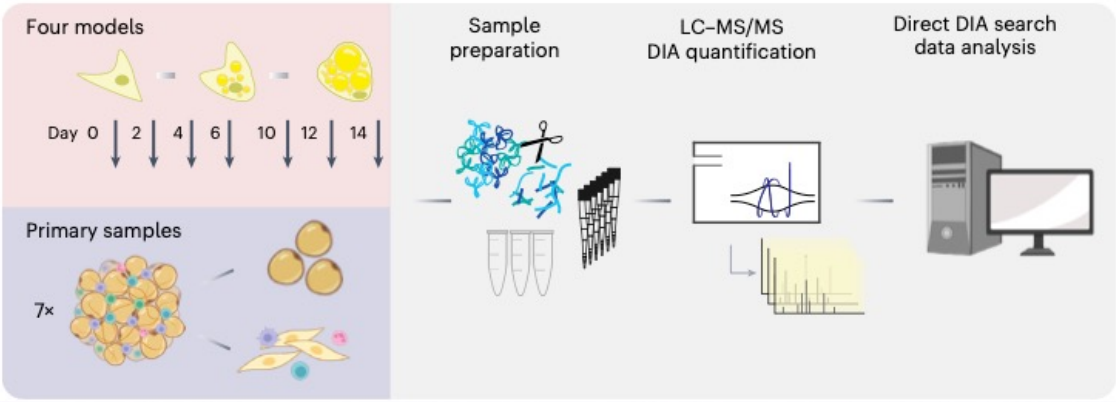
DIA LC–MS
Temporal proteome

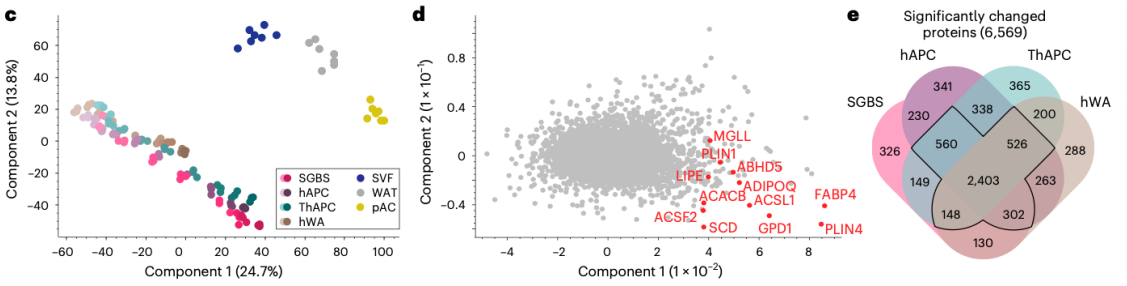
Organelle map
Gradient + SVM

Test function
C19orf12 perturbation

Klingelhuber et al. | NATURE METABOLISM (2024) | https://doi.org/10.1038/s42255-024-01025-8

Fig. 1 | Mapping the temporally resolved core proteome of human adipogenesis.





The temporally resolved core proteome of human adipogenesis

To define the core proteome trajectory during human adipogenesis, we performed liquid chromatography–mass spectrometry (LC–MS) proteomics over the time course of differentiation across different human adipocyte models (Fig. 1a). All models are derived from human adipocyte precursor cells (hAPCs) isolated from the stromal vascular fraction (SVF) of WAT and have the capacity to differentiate into adipocytes upon treatment with pro-adipogenic cocktails. Two cell types were non-immortalized (Simpson–Golabi–Behmel syndrome (SGBS)[10] and hAPC[11,12]), whereas two were immortalized (TERT-hAPC[13] and hWA[14]). As hAPC and TERT-hAPC are from the same donor, these cells allowed us to control for the potential effects of the immortalization process. Thus, a total of four human model systems were included in this study to map the conserved proteomic landscape of adipogenesis independent of cell type-specific effects. Information regarding the origins, immortalization procedures and differentiation protocols is summarized in Supplementary Tables 1–3. For comparison, we also included primary samples of subcutaneous abdominal mature adipocytes (pACs), SVF (which contains immature adipocyte precursors)

Fig. 1 | Mapping the temporally resolved core proteome of human adipogenesis.

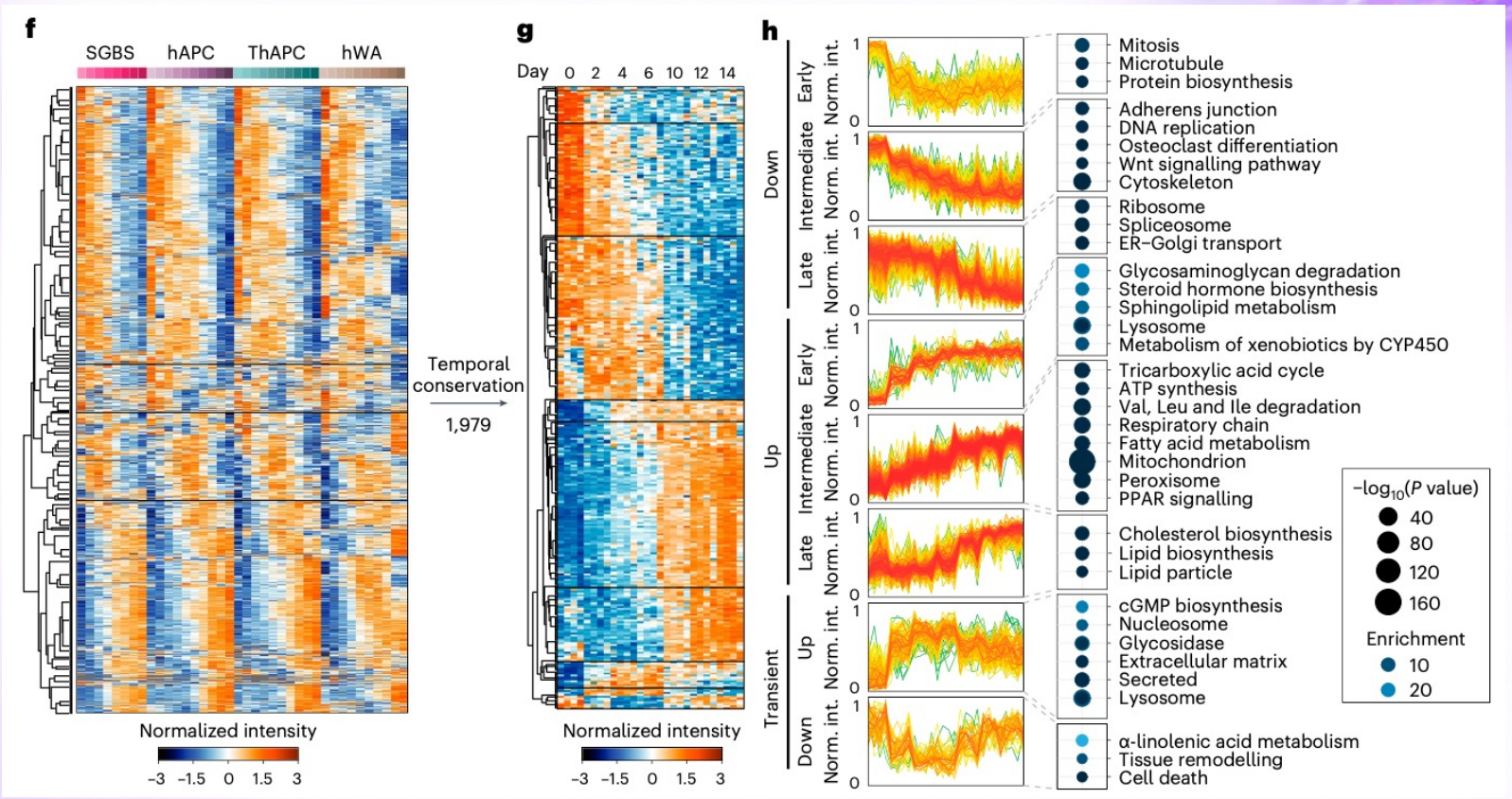
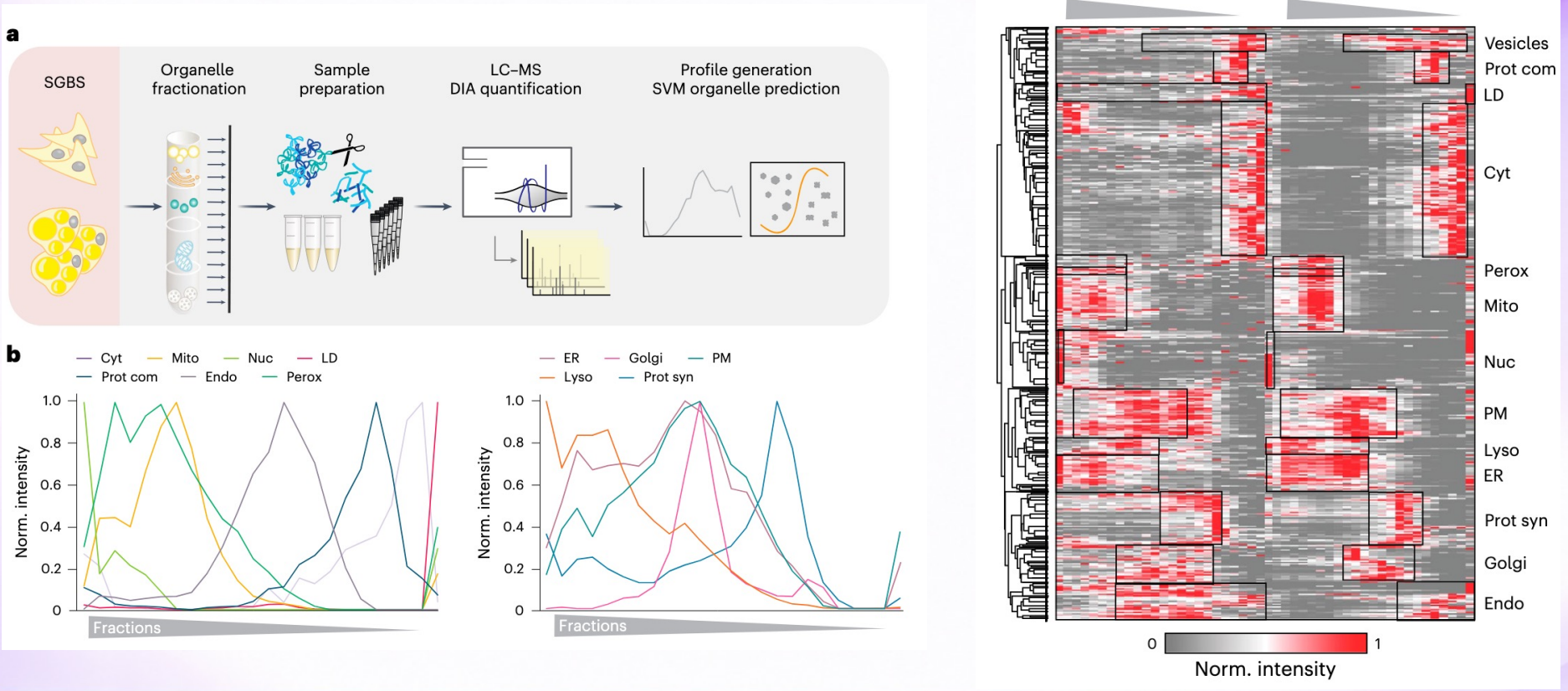


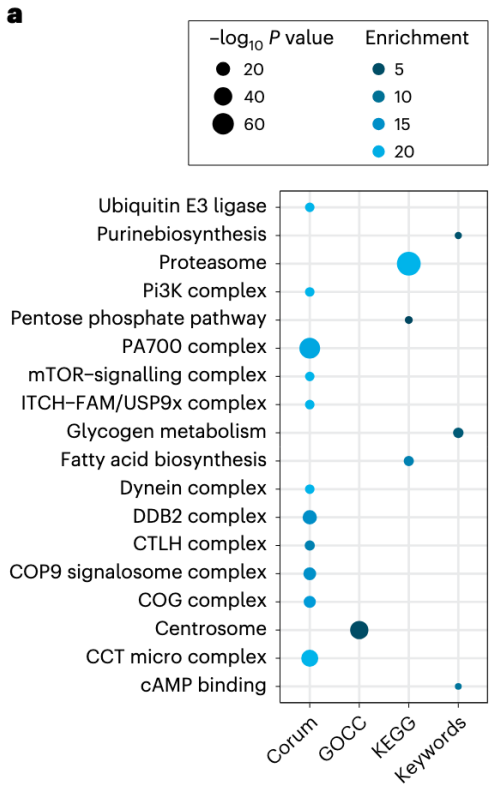
Fig. 2 | Generation of a cellular map of human adipogenesis.

Cellular map of human adipogenesis



Klingelhuber et al. | NATURE METABOLISM (2024) | https://doi.org/10.1038/s42255-024-01025-8

Fig. 5 | Spatial organization of lipid metabolism in adipocytes.



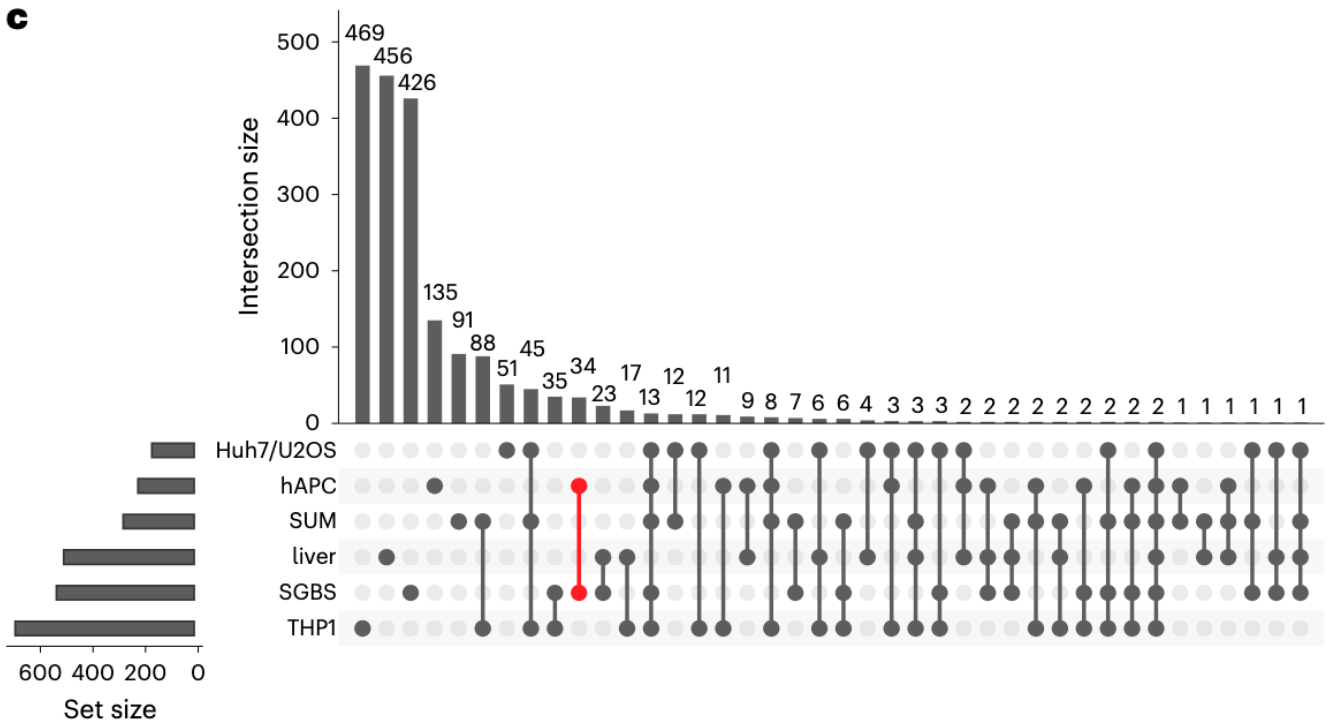
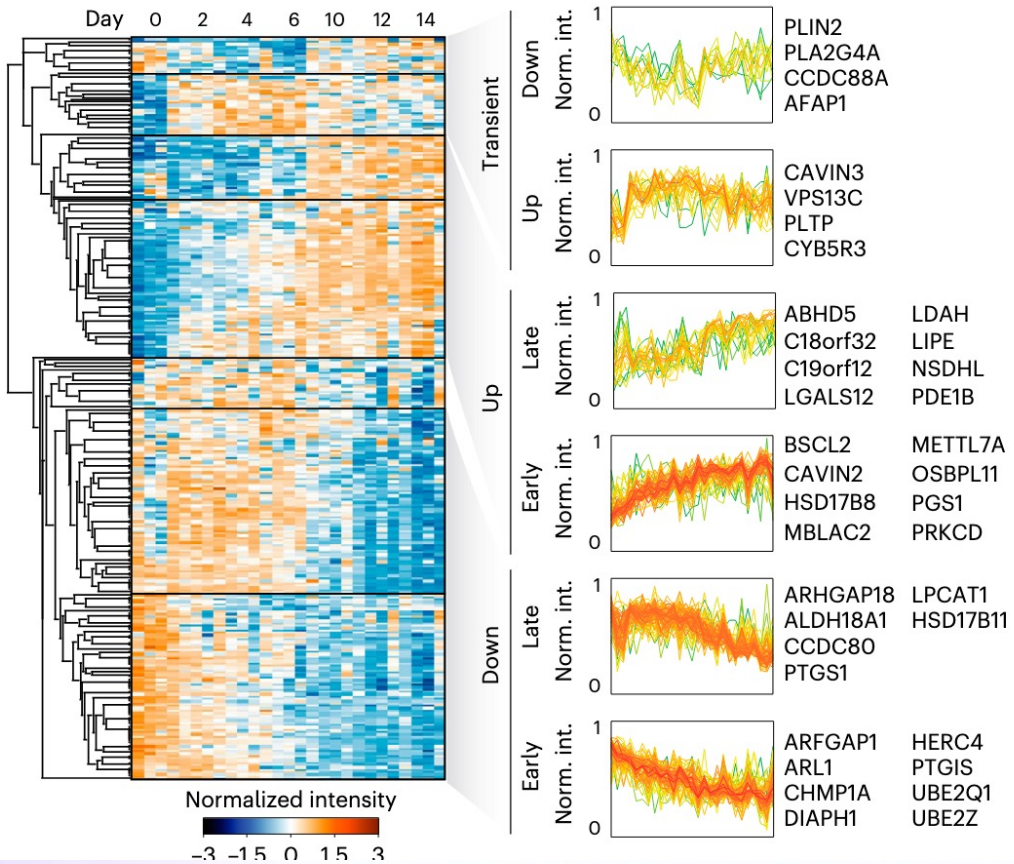
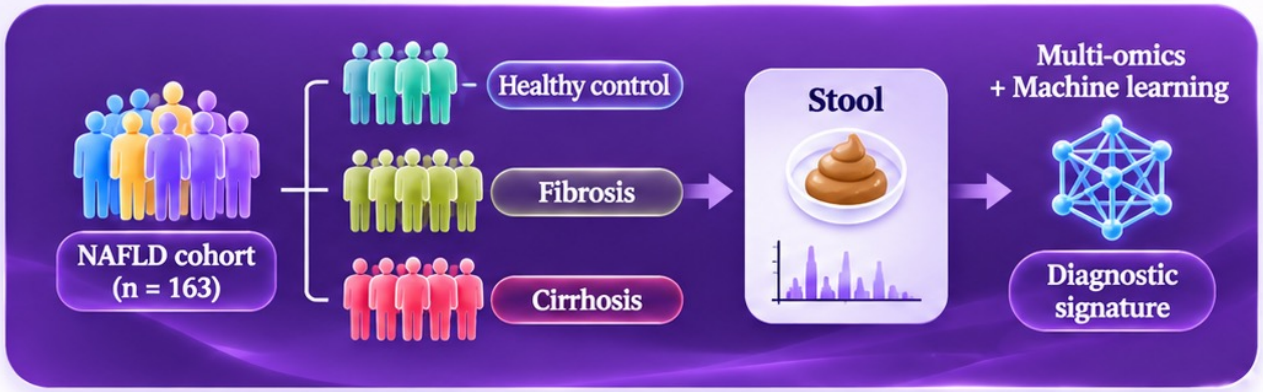


Fig. 5 | Spatial organization of lipid metabolism in adipocytes.



PAPER 2 · CELL METABOLISM · 2020

A Universal Gut-Microbiome-Derived Signature Predicts Cirrhosis



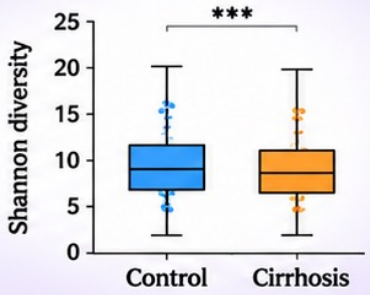
Study Overview

Question: Can gut microbiome signatures diagnose NAFLD-related cirrhosis?

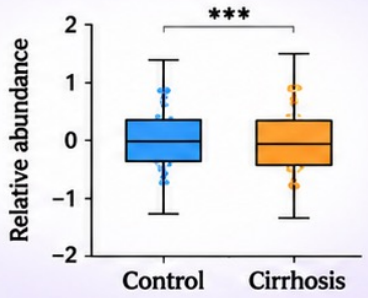
Approach: Stool metagenomics and metabolomics with machine learning.

Key finding: A universal stool-based signature combined with serum AST accurately distinguishes cirrhosis.

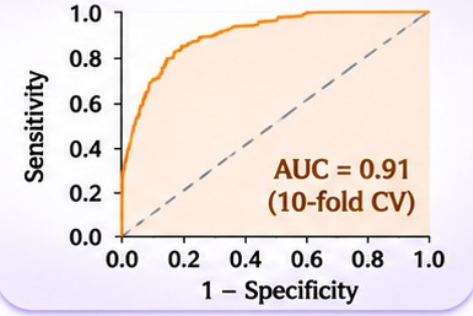
Microbiome features



Metabolite features



Model performance



Validation



Racially and geographically independent cohorts

Oh et al. | CELL METABOLISM (2020) | https://doi.org/10.1016/j.cmet.2020.06.005

Paper 2 · CELL METABOLISM · 2020

Paper 2: Key Findings



A Universal Gut-Microbiome-Derived Signature Predicts Cirrhosis



Study Overview

Cohort: 163 participants (non-NAFLD controls, NAFLD-cirrhosis, and first-degree relatives)

Approach: Stool metagenomics + metabolomics (LC-MS/MS)

Goal: Identify and validate diagnostic signatures for NAFLD-cirrhosis

Oh et al. | 2020

HIGHLIGHTS — paraphrased from the original



Non-invasive detection of cirrhosis by using microbial species, age, and serum measures.



Stool microbial and metabolite signatures independently predict NAFLD-cirrhosis.



Machine-learning-based prediction of cirrhosis validated in independent cohorts.

Oh et al. | CELL METABOLISM (2020) | https://doi.org/10.1016/j.cmet.2020.06.005

Figure 1. Major alterations in the gut microbiome profiles of the NAFLD-cirrhosis group compared with the Non-NAFLD control group

Table 1. Baseline Characteristics between (1) non-NAFLD Control Group and (2) NAFLD-Cirrhosis Group

Characteristics	Non-NAFLD Group (N = 54) (1) (SD)	NAFLD-Cirrhosis Group (N = 27) (2) (SD)	Wilcoxon p Value (1 ↔ 2)
Demographics			
Age (years)	45.85 (19.86)	64.74 (9.80)	<0.001
Female (n%)	39 (72.2%)	22 (81.5%)	
White (n%)	43 (79.6%)	9 (33.3%)	
Hispanic (n%)	8 (14.8%)	15 (55.6%)	
BMI (kg/m^2)	26.07 (6.83)	32.85 (10.06)	<0.001
Clinical			
Type 2 Diabetes (n%)	1 (1.9%)	23 (85.2%)	
Biological Data			
AST (U/L)	22.31 (8.44)	46.26 (23.11)	<0.001
ALT (U/L)	18.81 (8.84)	44.3 (36.86)	<0.001
Alk P (U/L)	68.17 (19.04)	112.19 (39.63)	<0.001
GGT (Ui/L)	17.63 (8.02)	97.04 (57.71)	<0.001
Total bilirubin (mg/dL)	0.51 (0.25)	2.33 (6.63)	<0.01
Direct bilirubin (mg/dL)	0.12 (0.04)	1.36 (5.21)	<0.001
Albumin (g/dL)	4.53 (0.33)	4.06 (0.61)	<0.001
Glucose (mg/dL)	86.09 (9.04)	132.07 (61.85)	<0.001
Hemoglobin A1c (%)	5.65 (0.32)	7.06 (1.88)	<0.001
Insulin (U/mL)	8.79 (6.35)	43.52 (33.06)	<0.001
Triglycerides (mg/dL)	81.50 (44.73)	147.42 (136.45)	<0.001
Total cholesterol (mg/dL)	188.57 (40.94)	162.83 (64.23)	<0.05
HDL-cholesterol (mg/dL)	69.87 (19.89)	49.17 (18.49)	<0.001
LDL-cholesterol (mg/dL)	102.43 (32.17)	79.3 (32.42)	<0.01
Platelet count (10^3/μL)	246.94 (48.54)	149.11 (72.73)	<0.001
Prothrombin time	10.56 (0.69)	13.04 (3.66)	<0.001
INR	1.02 (0.07)	1.22 (0.35)	<0.01
Ferritin (ng/mL)	103.89 (82.9)	132.79 (123.19)	n.s.
Imaging Data			
MRI-PDFF (%)	2.41 (0.86)	5.16 (4.11)	<0.01
MRE (kPa)	2.13 (0.37)	5.04 (2.68)	<0.001

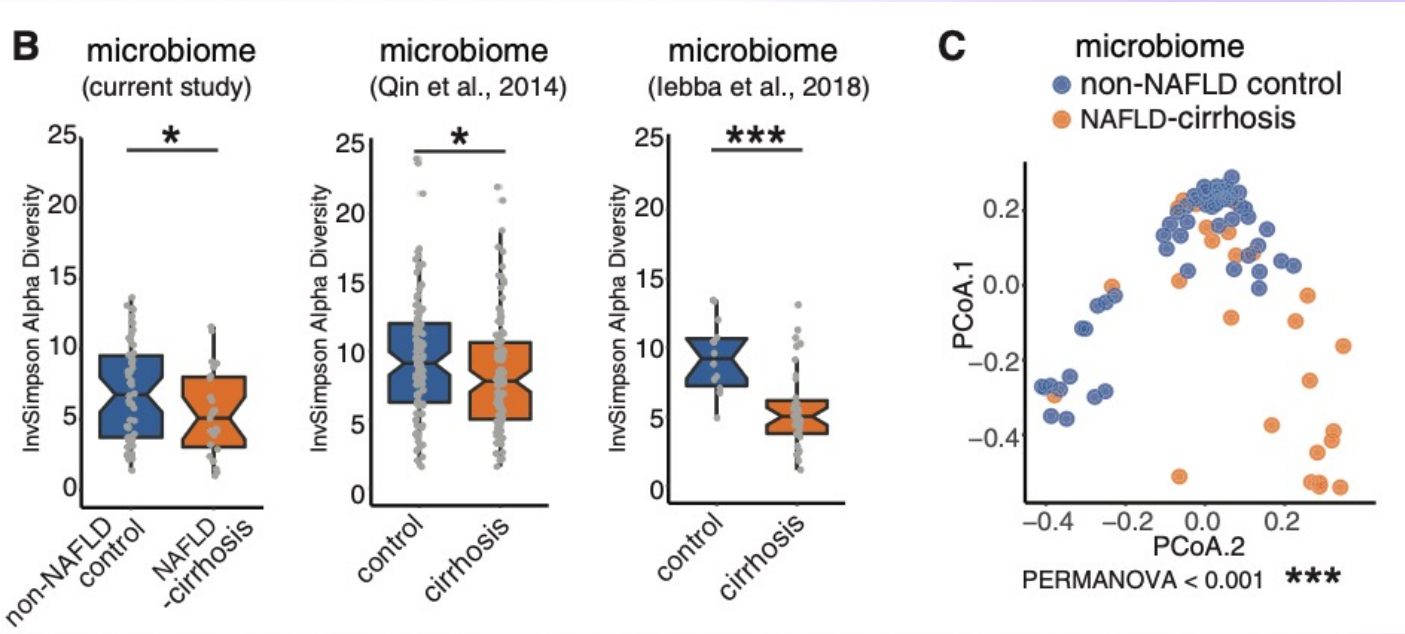


Figure 1. Major alterations in the gut microbiome profiles of the NAFLD-cirrhosis group compared with the Non-NAFLD control group

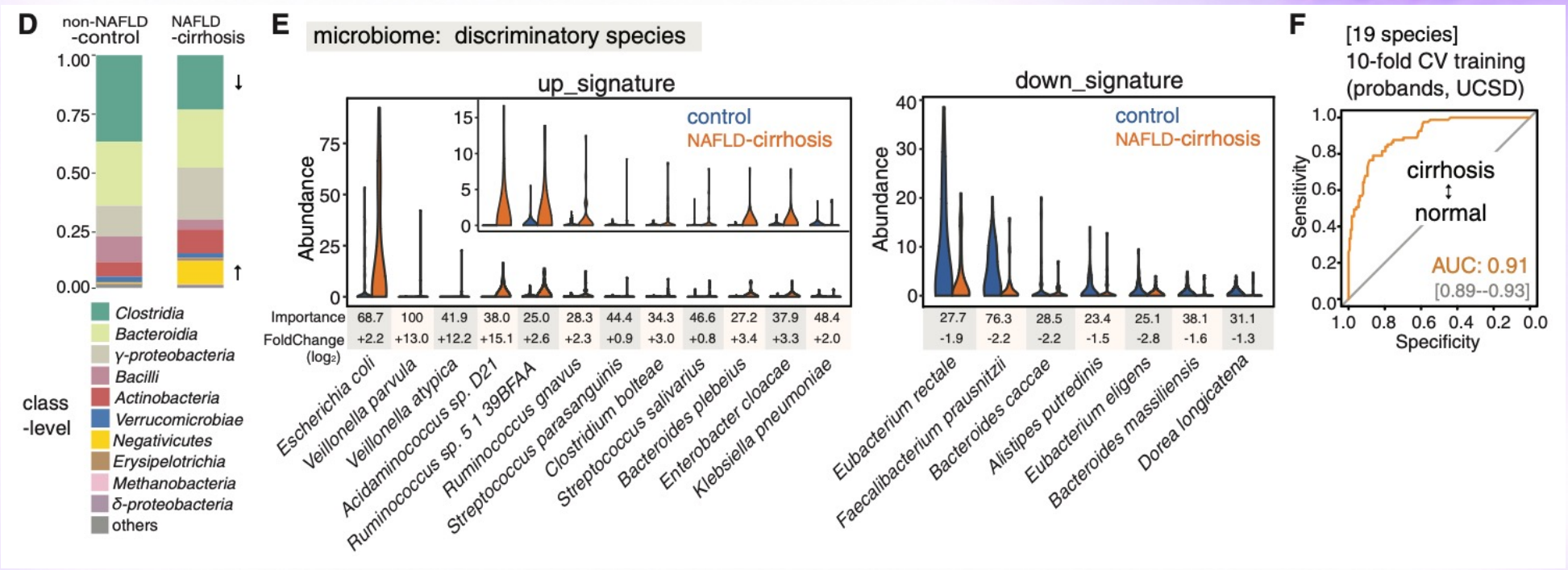


Figure 3. Identification of discriminatory metabolites for cirrhosis

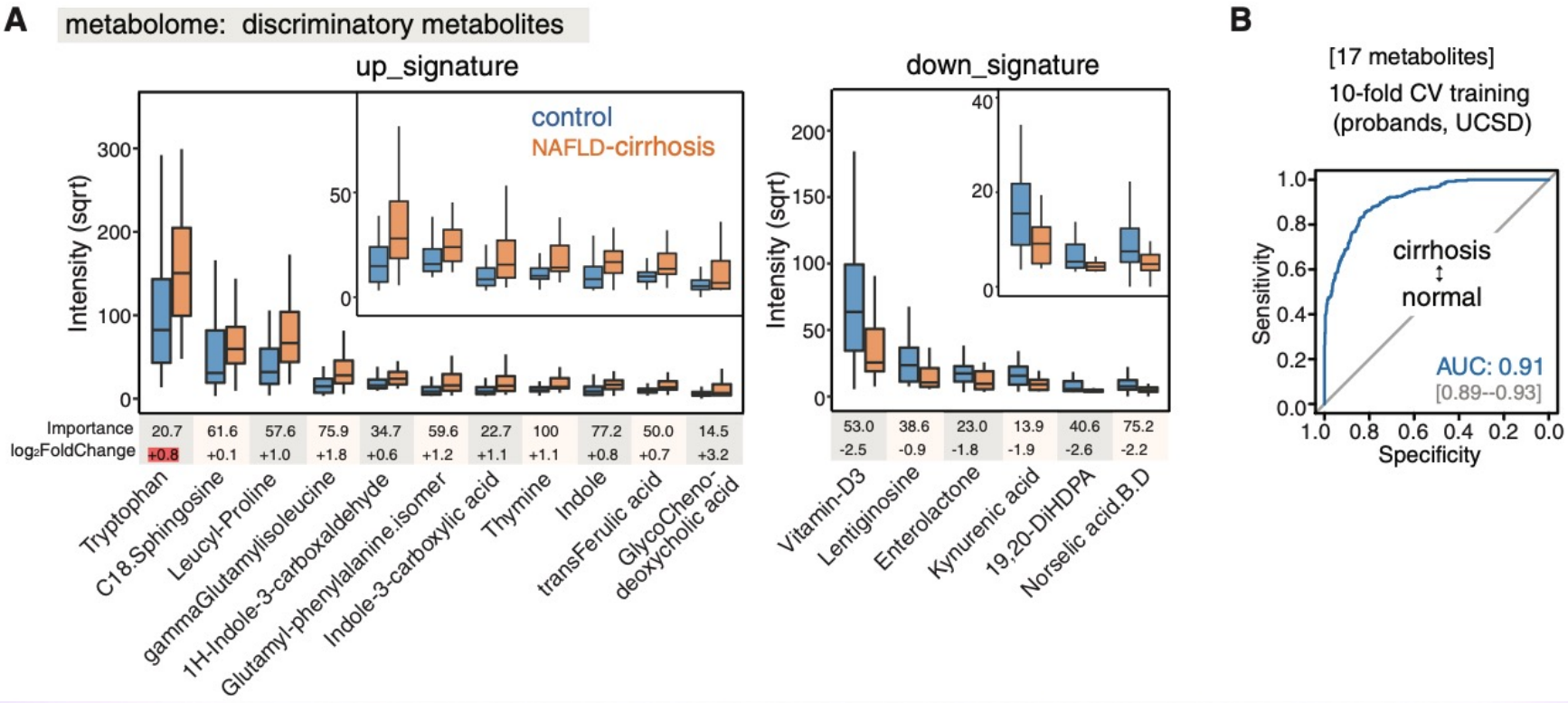


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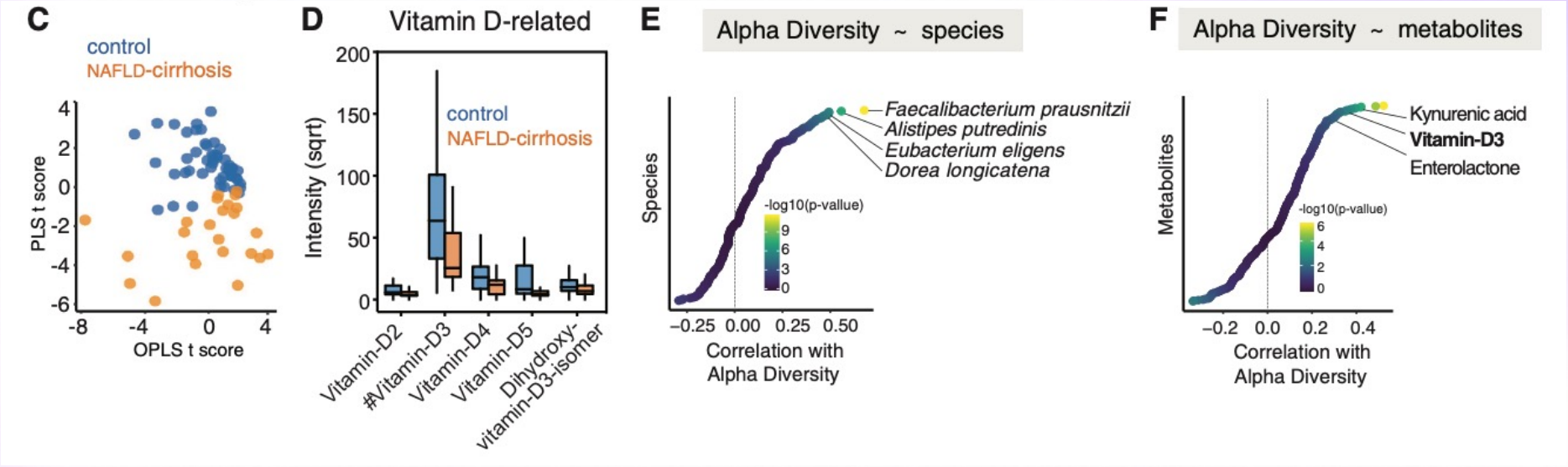


Figure 4. Interaction network of metagenomic and metabolomic features for cirrhosis

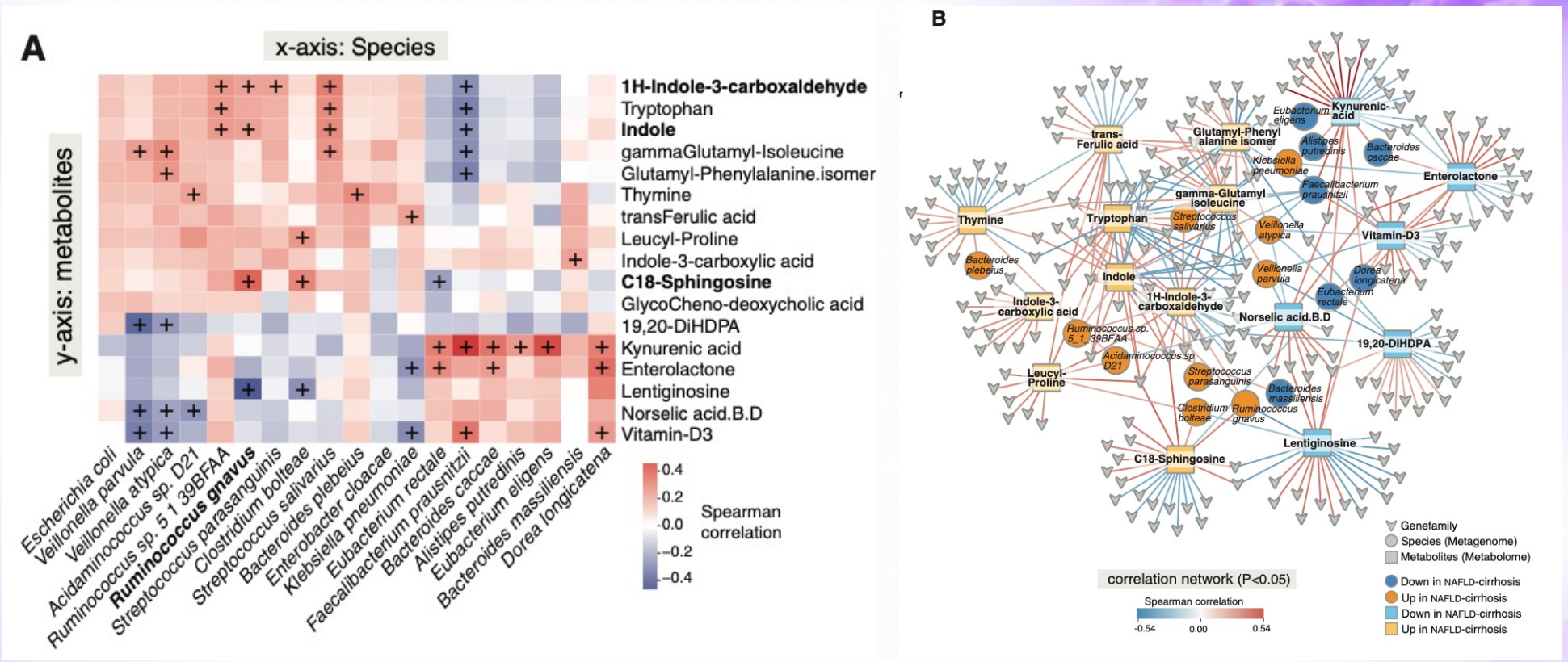
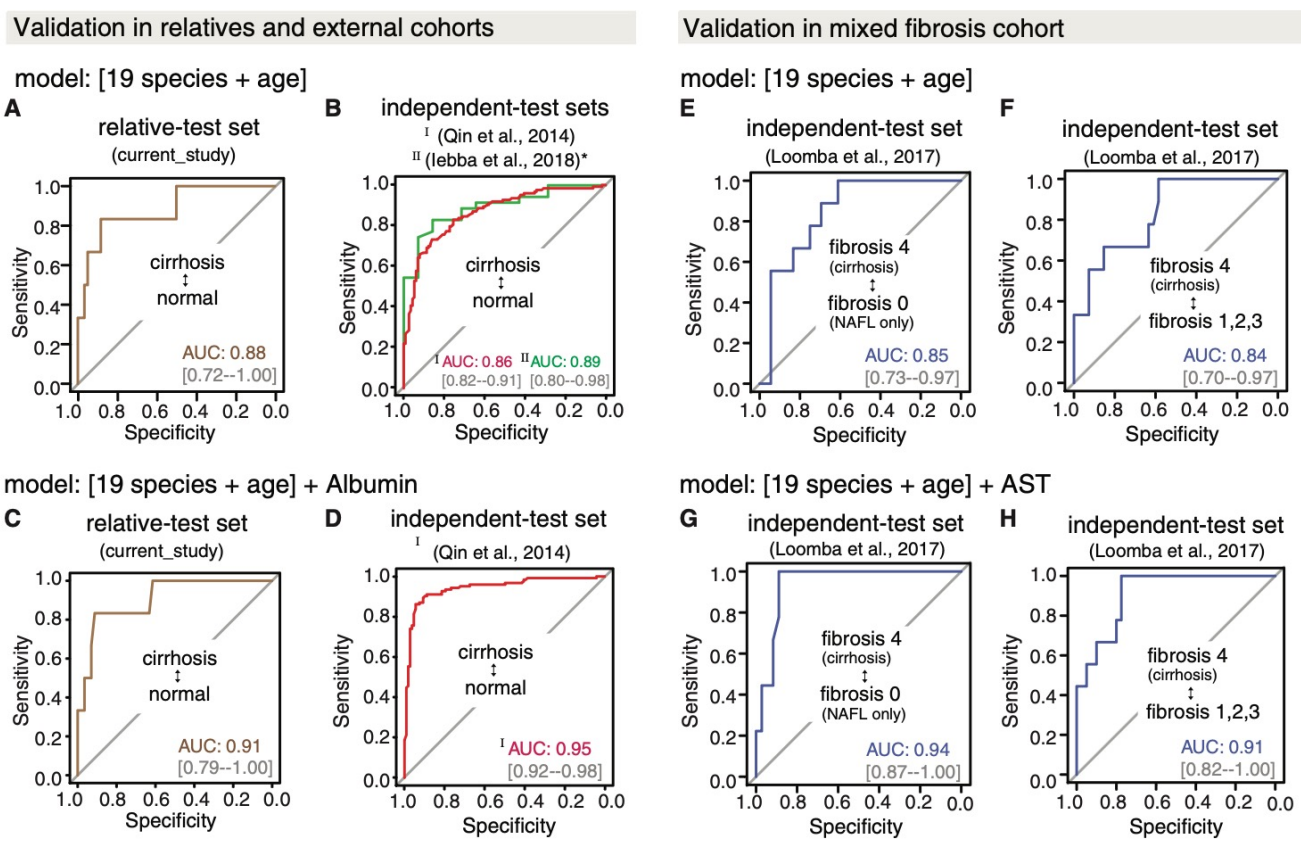
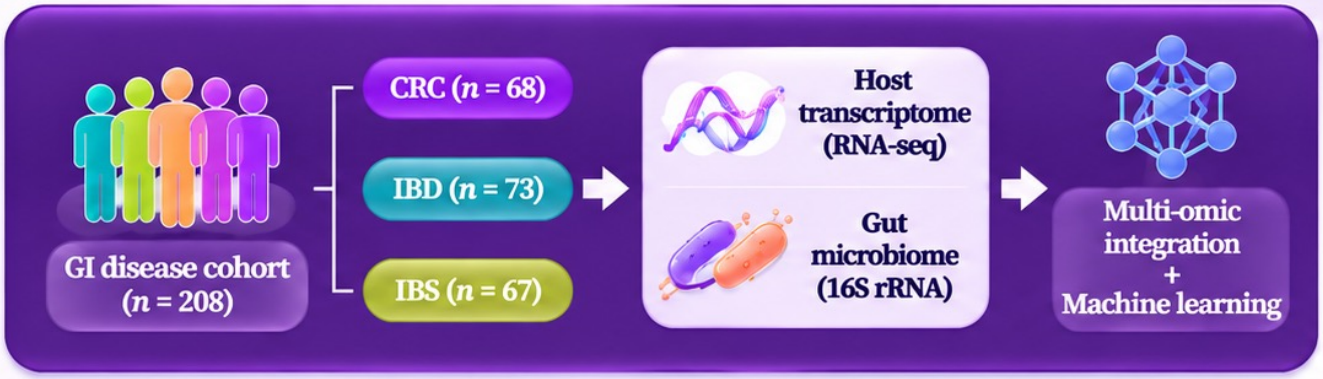


Figure 5. Validation of the Machine Learning RF Model from Stool Metagenome for Cirrhosis



Paper 3 · NATURE MICROBIOLOGY · 2022

Paper 3: Study Overview



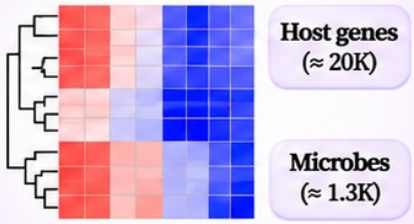
Study Overview

Aim: Jointly characterize host gene–microbiome associations in CRC, IBD and IBS.

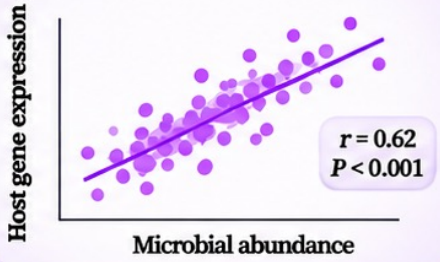
Approach: Multi-omic integration of paired host transcriptome and gut microbiome (n = 208) with machine learning.

Key finding: Shared and disease-specific host gene–microbiome pathways across GI diseases.

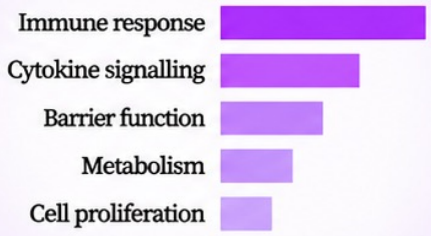
Multi-omic data



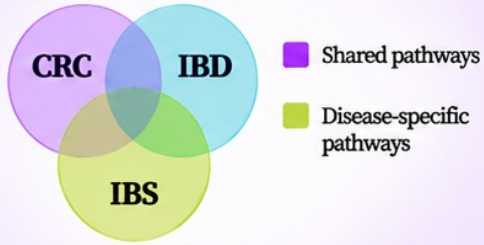
Microbe–gene associations



Functional pathways



Disease patterns



Priya et al. | NATURE MICROBIOLOGY (2022) | https://doi.org/10.1038/s41564-022-01121-z

Paper 3 · NATURE MICROBIOLOGY · 2022

Paper 3: Research Question

Identification of shared and disease-specific host gene–microbiome associations across human diseases



Study Overview

Question: How do host gene expression and gut microbiome interact in CRC, IBD and IBS?

Approach: Integrative multi-omic analysis with machine learning on paired RNA-seq and 16S data.

Key finding: Identification of shared and disease-specific host gene–microbiome associations and related biological pathways.

Priya et al. | NATURE MICROBIOLOGY (2022) | https://doi.org/10.1038/s41564-022-01121-z

Fig. 1 | Integrating host gene expression and gut microbiome abundance in CRC, IBD and IBS.

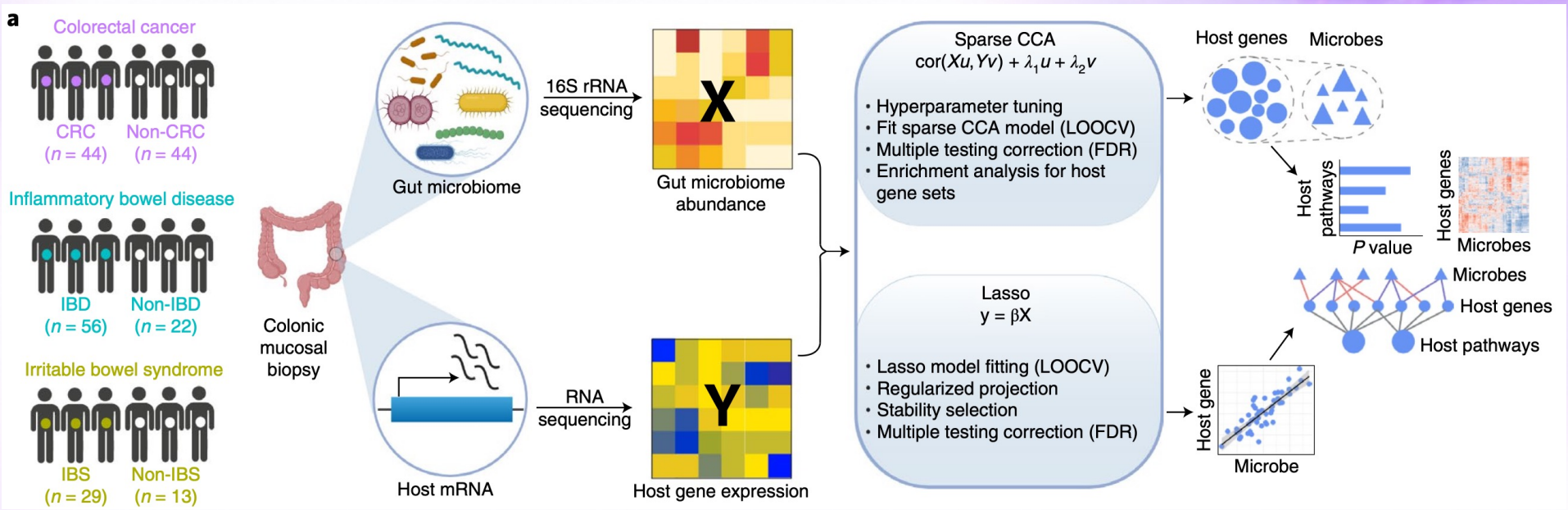
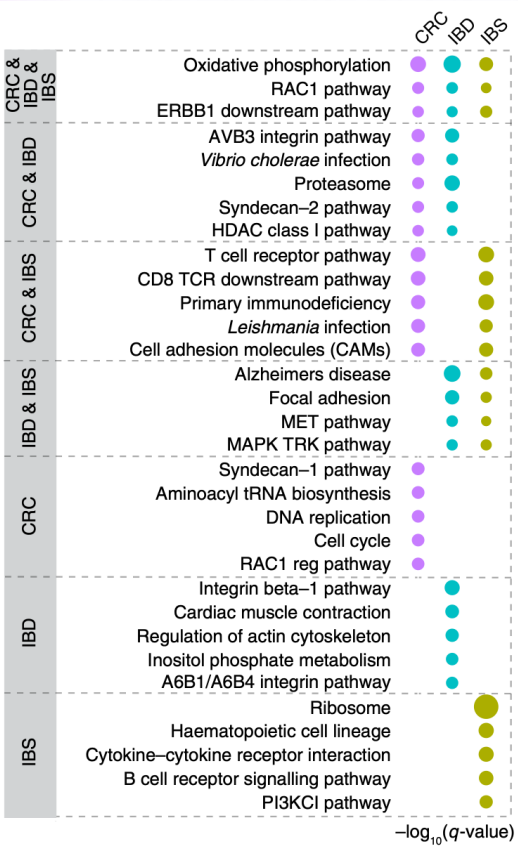


Fig. 2 | Shared immunoregulatory and metabolic host pathways associate with disease-specific gut microbes across human diseases.



Supplementary Tables 2–4). We then performed pathway enrichment analysis on the set of host genes in each significant component to determine host pathways that associate with gut microbes in a disease. We identified 'shared' pathways, namely host pathways for which gene expression correlates with gut microbes across disease cohorts, and 'disease-specific' pathways, namely host pathways for which gene expression correlates with gut microbes in only one of the three disease cohorts (Fig. 2a, Fisher's exact test, Benjamini-Hochberg FDR < 0.1; Supplementary Table 5). For simplicity, we focused on the top five most significant shared and disease-specific pathways (Fig. 2a). We found three pathways shared

Fig. 2 | Shared immunoregulatory and metabolic host pathways associate with disease-specific gut microbes across human diseases.

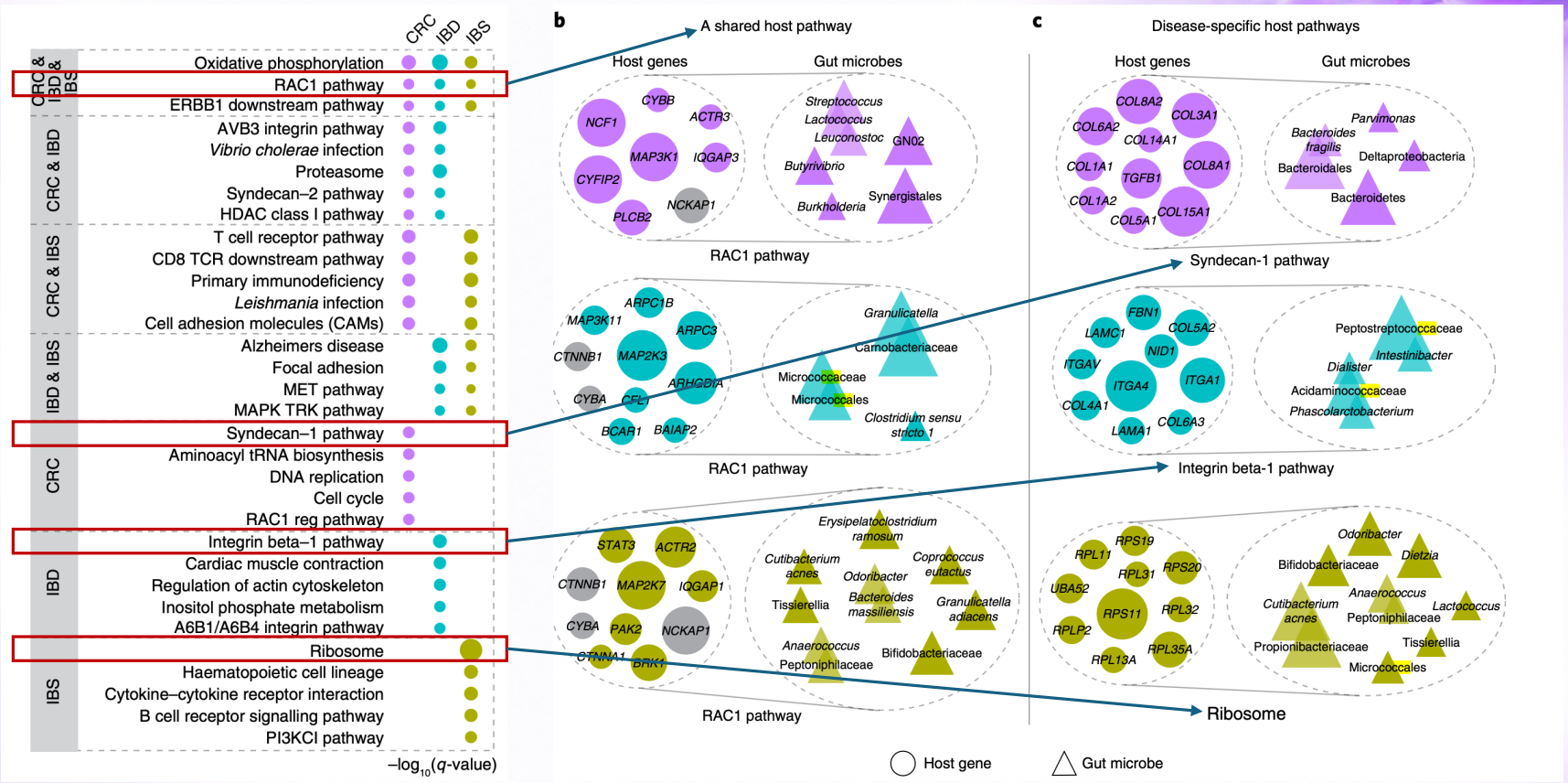
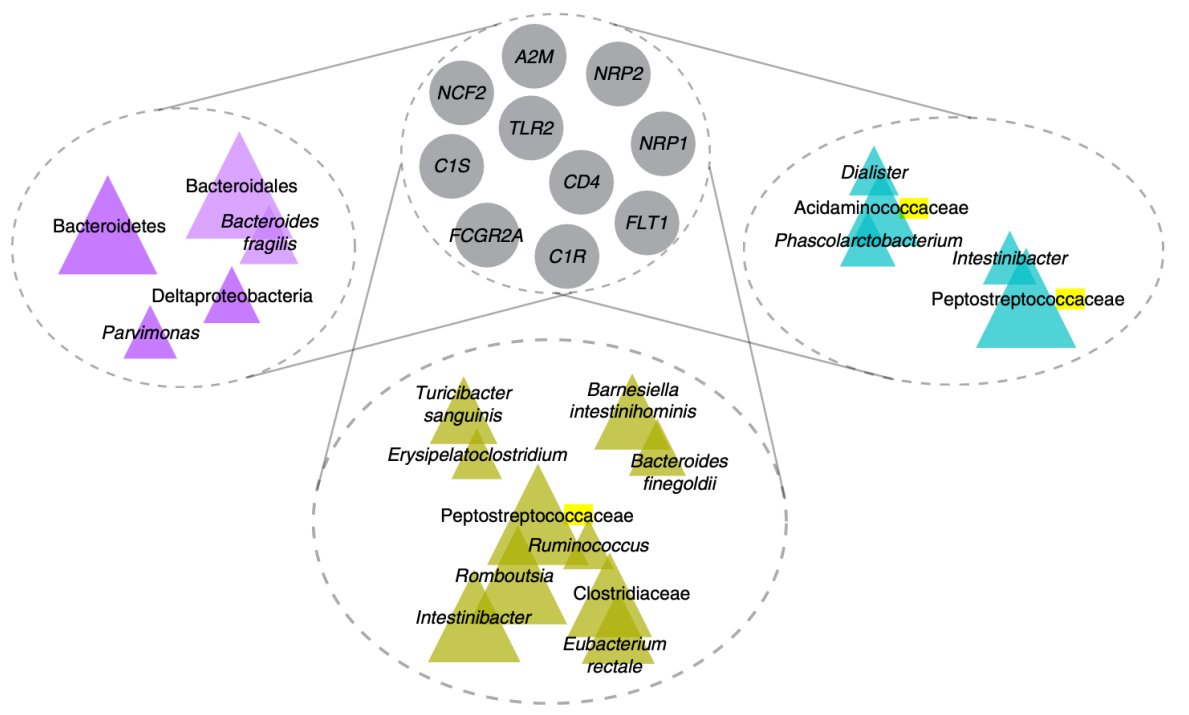
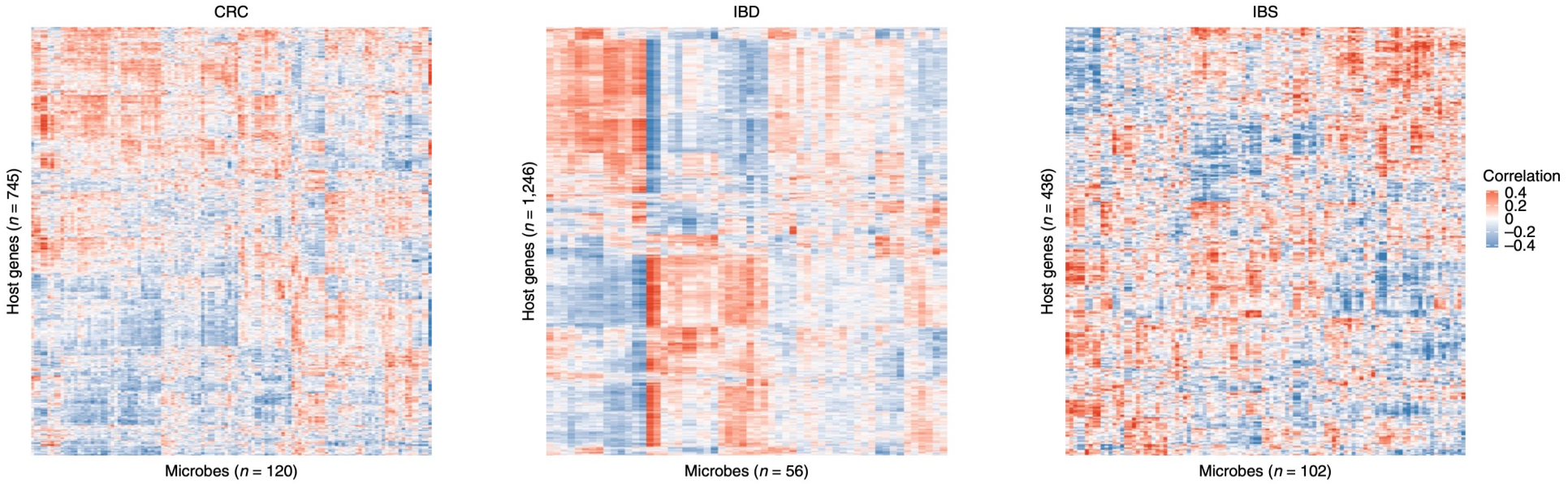


Fig. 2 | Shared immunoregulatory and metabolic host pathways associate with disease-specific gut microbes across human diseases.



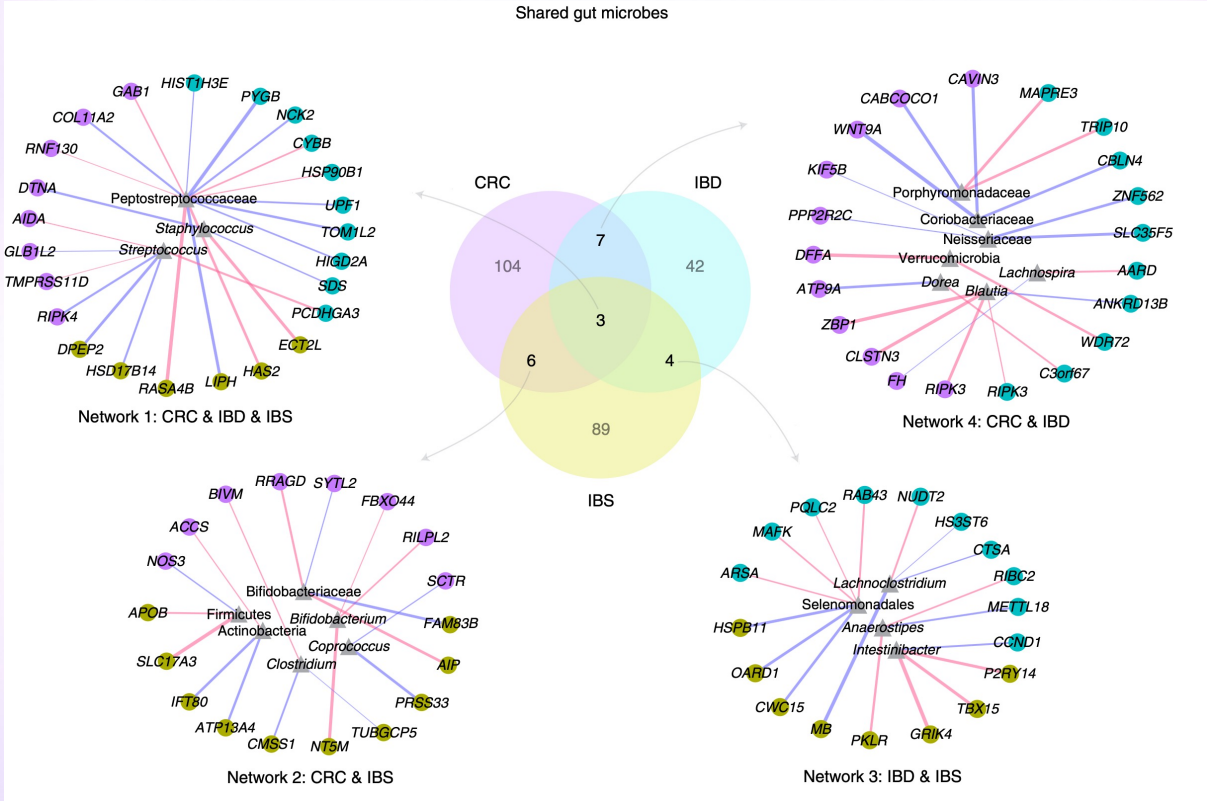
A common set of host genes (grey circles) that are associated with disease-specific sets of microbes. These host genes are enriched for immunoregulatory and acute inflammatory response pathways.

Fig. 3 | Specific gut microbes are associated with individual host genes and pathways in each disease.



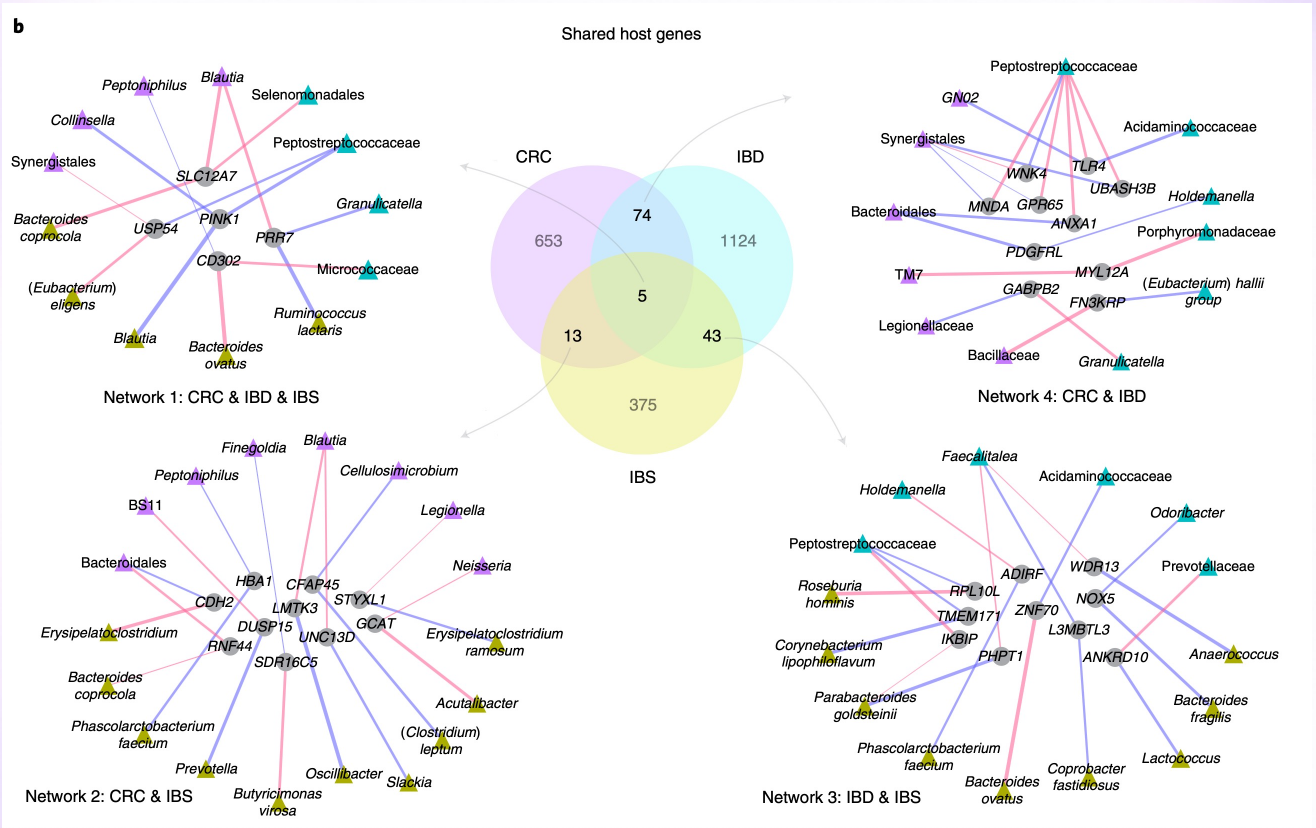
Heatmap showing the overall pattern of correlation between significant and stability-selected host genes (rows) and gut microbial taxa (columns) identified by the lasso model in CRC, IBD and IBS (FDR < 0.1).

Fig. 4 | Disease-specific gut microbe–host gene crosstalk.



Associations for ‘shared’ gut microbes, namely microbes that are associated with host genes in at least two diseases.

Fig. 4 | Disease-specific gut microbe–host gene crosstalk.

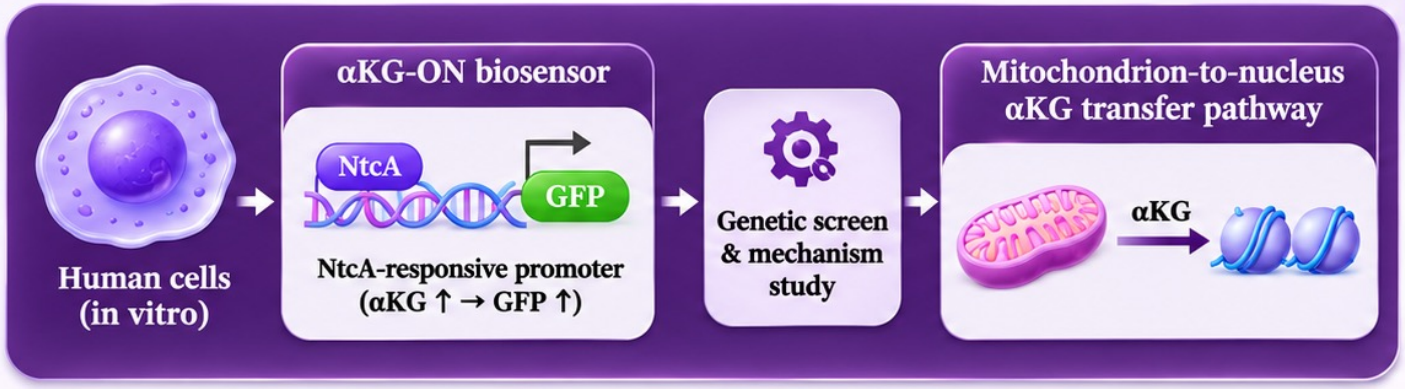


Associations for ‘shared’ host genes, that is, genes that are associated with microbes in at least two diseases.

Paper 4 · SCIENCE · 2026

Paper 4: Study Overview





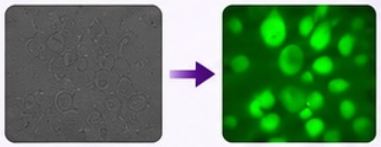
Study Overview

Aim: Define the mechanism by which αKG regulates chromatin via interorganelle metabolism.

Approach: Develop a transcriptional αKG biosensor, perform genetic screen, and validate in cells and mice.

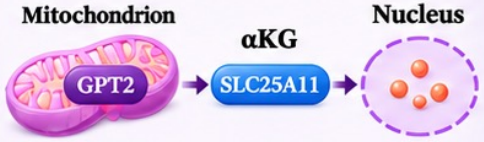
Key finding: A GPT2–SLC25A11 pathway supplies nuclear αKG to sustain chromatin demethylation.

αKG-ON biosensor



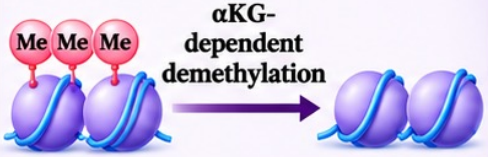
Low αKG (GFP low) → High αKG (GFP high)

Interorganelle pathway



GPT2 synthesizes αKG and SLC25A11 exports it to the nucleocytosolic compartment.

Chromatin regulation



In vivo validation



Gpt2-null mouse

αKG supplementation

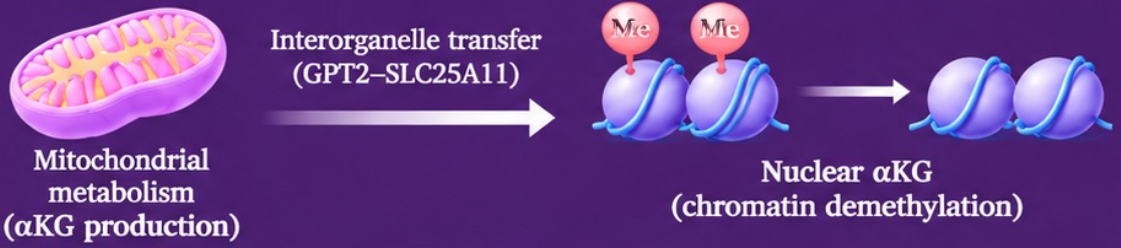
- Restores chromatin methylation
- Normalizes gene expression
- Improves mouse fitness

Sternisha et al. | SCIENCE (2026) | https://doi.org/10.1126/science.adx8675

Paper 4 · SCIENCE · 2026

Paper 4: Research Question

How is nuclear α-ketoglutarate (αKG) regulated to sustain chromatin demethylation and gene expression?



Key Questions

- **Mechanism:** How is αKG produced, transported, and maintained in the nucleus?
- **Function:** How does nuclear αKG control chromatin methylation and gene expression?
- **Disease relevance:** Does disruption of αKG metabolism cause epigenetic and developmental defects, and can it be rescued?

Mechanism: Source and transport of αKG



How is αKG synthesized and transferred from mitochondria to the nucleocytosol?

Function: Chromatin and gene regulation



How does nuclear αKG regulate chromatin methylation and gene expression?

Disease relevance: Pathogenesis and rescue



αKG supplementation

- Restores chromatin methylation
- Normalizes gene expression
- Improves mouse fitness

Does loss of αKG supply cause epigenetic and developmental defects, and can αKG supplementation rescue them?

Sternisha et al. | SCIENCE (2026) | https://doi.org/10.1126/science.adx8675

Paper 4

RESEARCH ARTICLE SUMMARY

CELL BIOLOGY

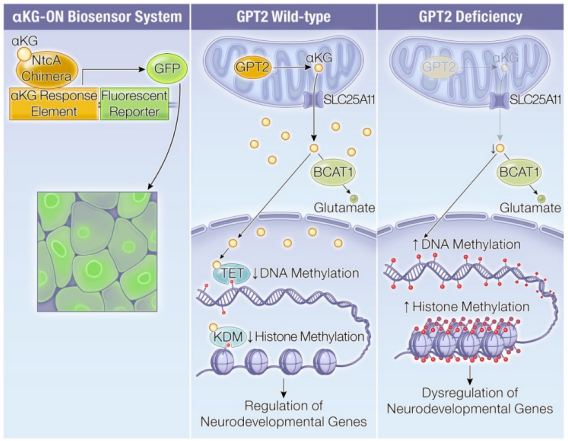
A transcriptional biosensor reveals mechanisms of α-ketoglutarate signaling to chromatin

Alex C. Sternisha†, Haocheng Li†, *et al.*

Full article and list of author affiliations: https://doi.org/10.1126/science.adx8675

INTRODUCTION: The metabolite α-ketoglutarate (αKG) is a substrate for dioxygenase enzymes that demethylate histones and DNA in the nucleus, influencing gene expression, cell fate, and tumor suppression. Despite the importance of nuclear αKG for chromatin-modifying enzyme function, mechanisms that govern this metabolite pool remain poorly defined. Revealing these mechanisms is vital for understanding how appropriate chromatin structure and gene expression patterns are maintained under homeostatic conditions and dysregulated in disease.

RATIONALE: Deciphering how nuclear αKG abundance is regulated in human cells has been challenging for several reasons. First, methods developed to study compartmentalized metabolism in mitochondria, lysosomes, and other organelles are not readily transferrable to nuclei. Second, intracellular αKG is regulated by a complex network of more than 100 enzymes and transporters that act directly on this metabolite. To address these challenges, we sought to create a biosensor capable of detecting changes in nuclear αKG and use it to identify the molecular processes that govern this metabolite pool.

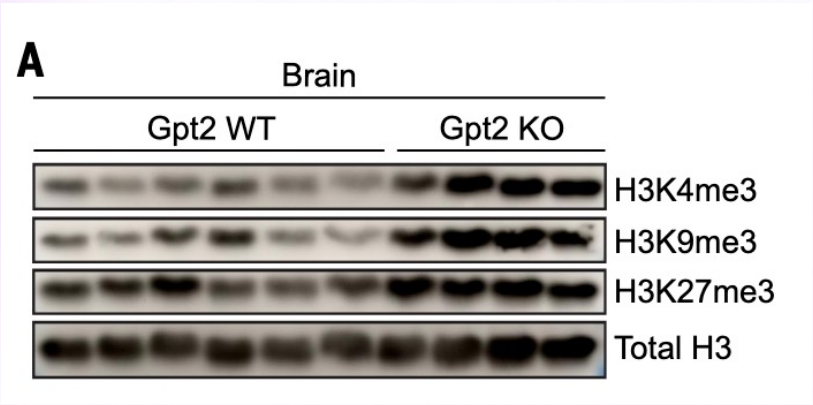


Central hypothesis: depletion of nuclear αKG and chromatin hypermethylation may contribute to disease phenotypes and transcriptional outcome

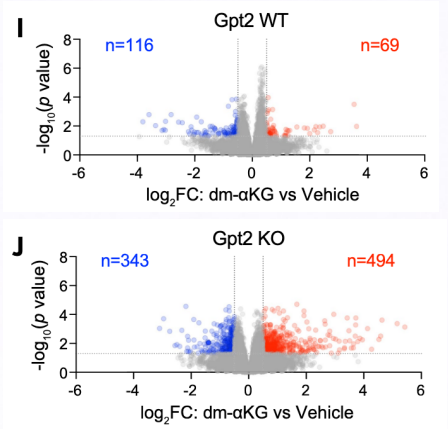
Epigenetic regulation: H3K4me3/H3K9me3/H3K27me3 ChIP-seq (Histone/Chromatin state, hypermethylation)
Transcriptional regulation: RNAseq

GPT2: mitochondrial glutamic-pyruvic transaminase 2 synthesizes αKG
SLC25A11: αKG exporter from mitochondria to cytosol
BCAT1: branched-chain amino acid transaminase 1 to import or catabolism to glutamate

Fig. 6. αKG-dependent neurodevelopmental gene dysregulation occurs in Gpt2 KO mice



Gpt2 is a critical enzyme involved in synthesis of αKG. Its deletion leads to upregulation of key histone methylation related genes and dysregulation of a cluster of key genes



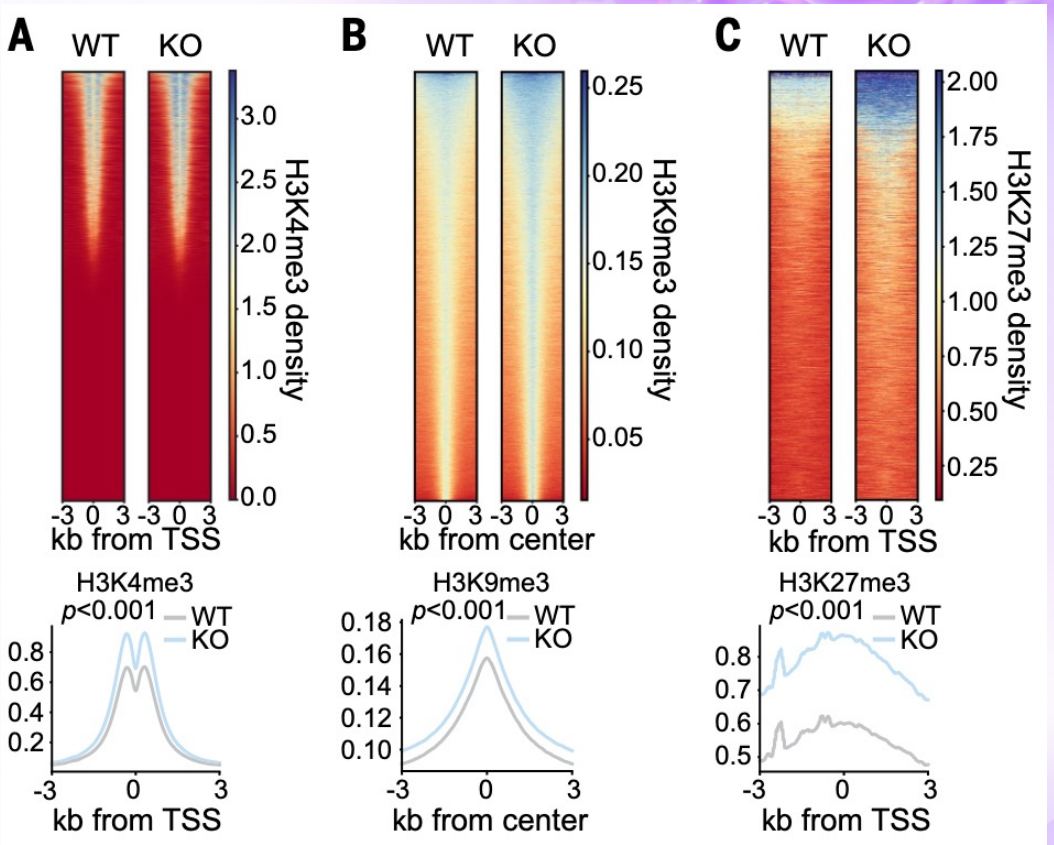
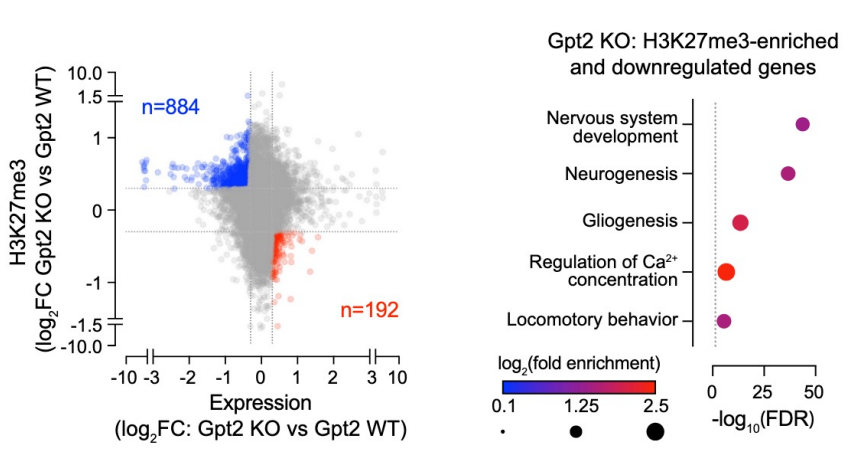
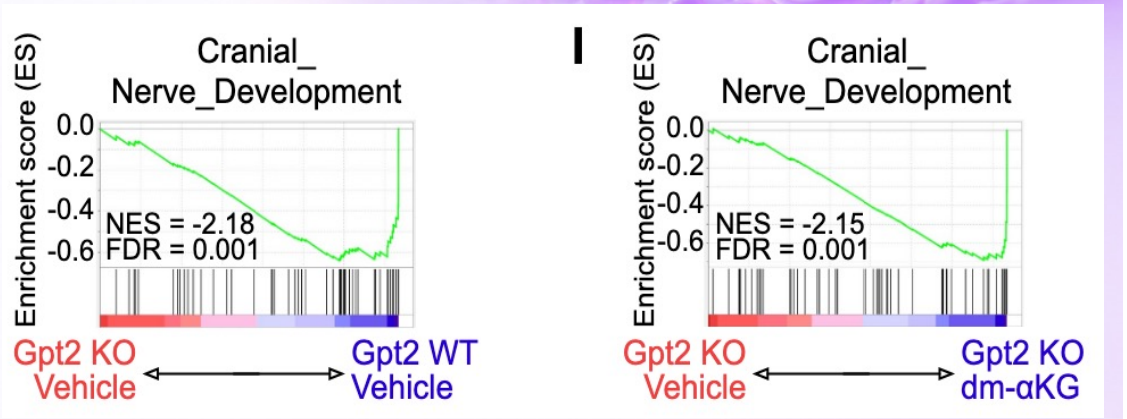
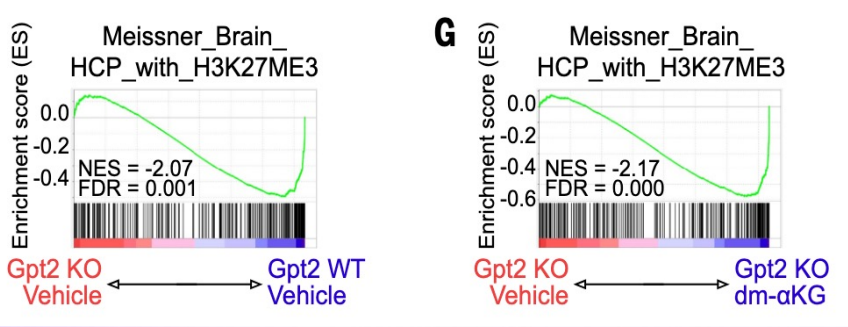
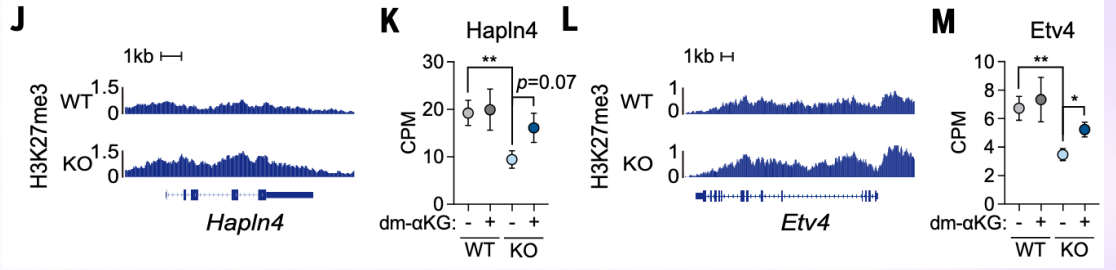


Fig. 6. αKG-dependent neurodevelopmental gene dysregulation occurs in Gpt2 KO mice









Sternisha et al. | SCIENCE (2026) | https://doi.org/10.1126/science.adx8675

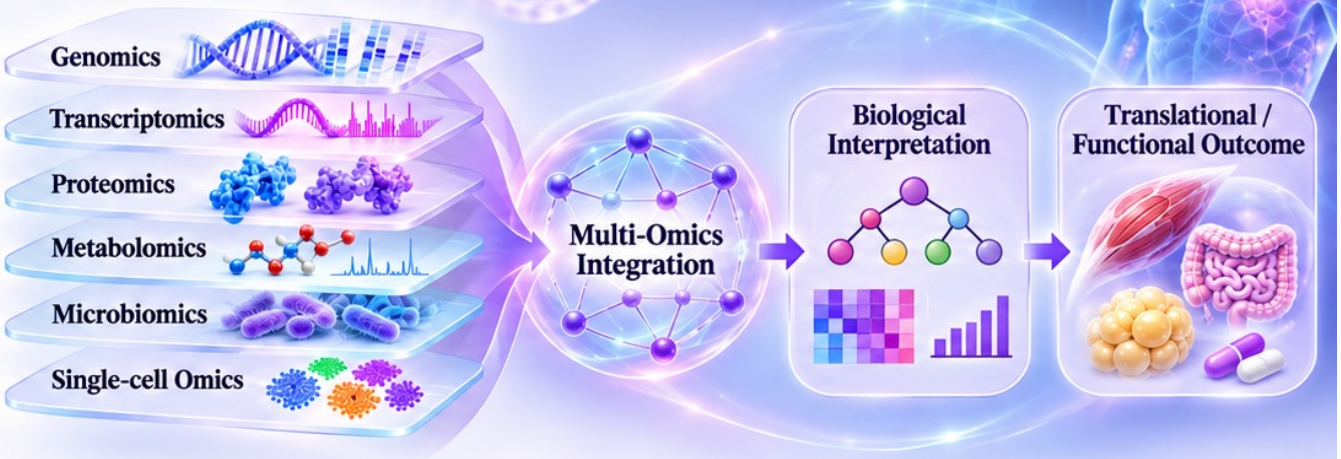
Week 11 | Bioinformatics: Multi-Omics Data Analysis

Key Takeaways

From Published Figures to Biological Discovery

Summary

- Start with a clear biological question and study design.
- Read original figures before accepting the paper's conclusion.
- Distinguish observation, statistical inference and mechanism.
- Integrate omics layers to identify robust pathways and regulators.
- Validate key findings experimentally whenever possible.



Student Checklist

- [x] What was measured?
- [x] What was compared?
- [x] What is statistically supported?
- [x] What still needs validation?

Question → Original Figures → Cross-Omics Integration → Biological Insight

Two 40-Minute Sessions | Fall 2026 | Dr. Liwei Xie | Shenzhen University of Advanced Technology

Questions & Discussion